

Funding Proposal

FP163: Sustainable Renewables Risk Mitigation Initiative (SRMI) Facility

Multiple Countries | International Bank for Reconstruction and Development and International Development Association (World Bank) | Decision B.28/04

6 April 2021



GREEN
CLIMATE
FUND

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Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

“FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]”

A. PROGRAMME SUMMARY			
A.1. Project or programme	Programme	A.2. Public or private sector	Public
A.3. Request for Proposals (RFP)	Not applicable		
A.4. Result area(s)	Check the applicable <u>GCF result area(s)</u> that the <u>overall</u> proposed project/programme targets. For each checked result area(s), indicate the estimated percentage of <u>GCF budget</u> devoted to it. The total of the percentages when summed should be 100%.		
	<u>Mitigation</u>: Reduced emissions from: ☒ Energy access and power generation: ☐ Low-emission transport: ☐ Buildings, cities, industries and appliances: ☐ Forestry and land use: <u>Adaptation</u>: Increased resilience of: ☐ Most vulnerable people, communities and regions: ☐ Health and well-being, and food and water security: ☐ Infrastructure and built environment: ☐ Ecosystem and ecosystem services:		<u>GCF contribution</u>: 100%
A.5. Expected mitigation impact	88,980,000 CO ₂ eq tons	A.6. Expected adaptation impact	
A.7. Total financing (GCF + co-finance)	USD 1,563,500,000	A.9. Project size	Large (Over USD 250 million)
A.8. Total GCF funding requested	USD 280,000,000		
A.10. Financial instruments requested for the GCF funding	☒ Grant USD 54,000,000 ☒ Reimbursable Grant USD 30,000,000 ☒ Loan USD 176,000,000 ☐ Results-based payment n/a ☒ Guarantee USD 20,000,000		
A.11. Implementation period	12 years	A.12. Total lifespan	25 years for solar and wind projects / 50 years + for grid infrastructure
A.13. Expected date of AE internal approval	February 2, 2021	A.14. ESS category	A
A.15. Has this FP been submitted as a CN before?	Yes ☐ No ☒	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☐ No ☒
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☒ No ☐

A.19. Complementarity and coherence	Yes ☒ No ☐
A.20. Executing Entity information¹	Republic of Botswana represented by the Ministry of Finance and acting through the Ministry of Energy and the Botswana Power Corporation (BPC – the State Utility) Central African Republic (CAR) represented by the Ministry of Economy, Planning and Cooperation and acting through the Ministry of Energy Democratic Republic of Congo (DRC) represented by the Ministry of Finance and acting through the Ministry of Energy Republic of Kenya represented by the National Treasury and acting through the Ministry of Energy Republic of Mali represented by the Ministry of Finance and acting through the Ministry of Energy, the Société de Gestion de l’Energie de Manantali (SOGEM), and <i>Electricité du Mali</i> (EDM – the State Utility) Republic of Namibia represented by the Ministry of Finance and acting through the Ministry of Energy and NamPower (the State Utility) Republic of Uzbekistan represented by the Ministry of Finance and acting through the Ministry of Energy
A.21. Executive summary (max. 750 words, approximately 1.5 pages)	
While the proportion of renewable energy (RE) generation is rising every year, it remains a small percentage of the total world-wide electricity production and far from the targets needed to meet the Paris Agreement objectives. The proportion of the world electricity generated by wind, solar – together Variable Renewable Energy (VRE), biomass and waste-to-energy, geothermal, marine and small hydro rose from 6 percent in 2010 to 12.9 percent in 2018. Large solar and wind energy deployment is needed for enhancing energy access and energy security and to stay below the Climate Change Paris Agreement 2C° scenario. According to the Sustainable Development Scenario of the International Energy Agency (World Energy Outlook 2018), 950 GW of solar PV and 580 GW of wind need to be installed by 2025 in developing countries. It represents another 690 GW solar and 330 GW wind installation from 2019’s level. The limited installed capacity of cost-competitive solar and wind in developing countries reveals important regulatory, structural, and technical constraints. The main constraints identified are (i) limited generation and transmission planning capacity, (ii) inadequate regulatory framework that constrains the mobilization of sustainable private investments, (iii) limited procurement capacity to select the Independent Power Producers (IPPs), (iv) limited financial viability of off-takers and (v) VRE grid integration challenges due to weak grids especially in Sub-Saharan Africa. The World Bank through its Energy Sector Management Assistance Program (ESMAP), developed the Sustainable Renewables Risk Mitigation Initiative (SRMI) to address these challenges and propose a comprehensive financial and technical support to countries. Launched in 2018 for the COP	

¹ For the purposes of the loans and grants, the Executing Entities (EEs) would be the Borrowers/Recipients of the loans/grants. For the purposes of the guarantees contemplated under Component 3.2, the EEs would be the private sector RE project companies.

24 under the leadership of the World Bank in partnership with the *Agence Française de Développement* (AFD), the International Renewable Energy Agency (IRENA) and the International Solar Alliance (ISA), SRMI – previously known as the Solar Risk Mitigation Initiative – aims to support countries in developing and implementing sustainable renewable energy programs that will be more conducive to private investments to reduce reliance on public sector financing while maximizing socio-economic benefits for the countries (including from women empowerment's perspective).

Under the context of the Covid-19 crisis, SRMI's unique, integrated and replicable approach can support countries unlock a green economic stimulus through targeted public investments in order to leverage private sector investments at scale. SRMI aims to offer development and climate financing for (i) technical assistance to help countries develop evidence-based VRE targets, implement a sustainable renewable energy program, and maintain robust procurement processes with transaction advisors; (ii) critical public investments to enable integration of VRE, finance solar/wind park infrastructure, and increase access to electricity; and (iii) risk mitigation instruments to cover residual risks perceived by private investors. This financial support for renewable energy development is needed more than ever in the current situation alongside government economic support measures (such as short-term job allowances, direct cash transfers to citizens or targeted liquidity support to small and medium size enterprises).

The SRMI Facility to be funded by the GCF aims to provide technical and financial support to seven initial countries (Phase 1), namely Botswana, Central African Republic (CAR), Democratic Republic of Congo (DRC), Kenya, Mali, Namibia and Uzbekistan. If in 2017 those countries represented 1.2 percent of the total global greenhouse gas (GHG) emissions, under a BAU scenario they would on average triple their emissions between 2017 and 2030. With international support, the seven countries are committed to contribute to GHG emissions reductions especially focusing on their energy sector which is the main contributing sector. However, the countries have now prioritized immediate emergency response to manage the current Covid-19 crisis. The question of how the economic recovery is designed remains crucial in shaping the long-term pathways for emissions and determining whether the NDC targets can be achieved. If governments do not roll out low carbon development strategies and policies in response to the coming economic crisis, emissions could rebound and even overshoot previously projected levels by 2030, despite lower economic growth in the period to 2030.

It is therefore critical in this context to support policymakers to develop green stimulus interventions with catalytic concessional climate financing and support a medium-term strategy. The SRMI Facility, through the seven projects, is expected to contribute to the shift to low-emission sustainable development pathways as follows:

- **Directly support the development of an estimated 2.5 GW of RE,** (i) 1.8 GW of solar PV, (ii) 400 MW of concentrated solar plants (CSP), and (iii) 300 MW of onshore wind. Of this 2.1 GW are expected to be built during the lifetime of the projects and 400 MW expected to be built after the end of the projects. To unlock the RE power plants expected to be built during the lifetime of the projects, around USD 1 billion in public investment of which USD 250 million from GCF would need to have been disbursed. These public investment projects will also enable the integration of another 1.9 GW of RE. The integration of other projects is enabled due to the VRE integration analysis and grid investments.
- **Directly reduce GHG emissions by an estimated 97.93 m metrics tons of CO₂,** more precisely 88.98m reduction for the 2.1 GW RE built under the projects over the 25 years of their operation, and 8.95m metric tons of CO₂ for 0.4 GW of RE power plants over the 25 years of their operation that are expected to be prepared under the SRMI Facility projects but not built during the lifetime of the projects.

- **Unlock an estimated USD 3.3 billion of private investments**, including USD 3 billion for the 2.1 GW built during the lifetime of the projects and USD 300 million for the 0.4 GW built after the end of the projects.
- **Provide access to clean low-emission electricity to an estimated 4.2 million people through the electrification investments.**
- **Strengthen institutional and regulatory systems for planning and development.**
- **Reduce locking-in long-lived, climate-vulnerable infrastructure by increasing resilience of grid infrastructure.**

The SRMI Facility aims to be scaled up to a total USD 400 million in GCF funding amount by including additional countries, namely Indonesia and Vietnam, with whom significant and extensive engagements have been had to date. These efforts are expected to result into NOLs from these two countries post submission of this funding proposal. Vietnam and Indonesia are expected to have projects that meet the agreed eligibility criteria as defined in the present funding proposal. The Phase 2 of the Facility that is to be complementary to the current Phase 1 aims to –based on current project information - (i) avoid a total of 286.32 m metric tons of CO₂, including 60.6 m metric tons of CO₂ during project lifetime and 225.7 m metrics tons built after the end of the project, (ii) unlock a total of 6.1 GW, including in the short-term 2 GW and in the medium-term 4.1 GW of RE, and (iii) leverage a total of USD 10.2 bl, including USD 1.2 bl during the project lifetime and USD 9 bl built after the end of the project. Once and if approved by GCF Board, Phase 2 will be complementary to the present Funding Proposal.

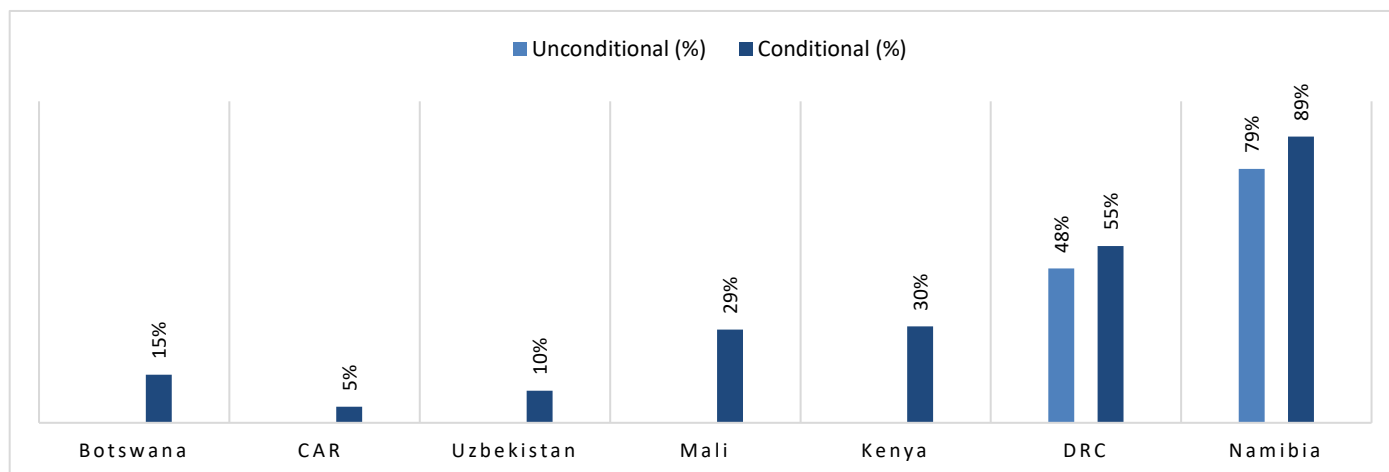
B. PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

The SRMI Facility aims at deploying the SRMI approach to unlock RE deployment by leveraging sustainable private investments starting with seven countries, namely Botswana, CAR, DRC, Kenya, Mali, Namibia and Uzbekistan. These countries were selected based on the willingness and interest of their respective governments to embrace renewable energy deployment and combat the challenges they face in the implementation of their NDCs and low-emissions development pathways. In addition, those countries represent a large sample of developing countries – with around 200 million inhabitants – facing different constraints, from very small to very large grids, low to high electrification rates, availability or not of domestic fossil fuels, and fragile and post conflict countries to medium income countries. The lessons learnt from the implementation of SRMI in those countries will enable to replicate the approach at scale in a sustainable and affordable manner, contributing to fast-track the Sustainable Development Goals (in particular SDG 13 on Climate Action and SDG 7 on Affordable and Clean Energy).

Under the context of the Covid-19 crisis, the SRMI Facility's projects aim to support countries in their energy transition and contribute to the overall mitigation agenda at a time where the economic recovery design will determine whether the Paris Agreement targets can be achieved. In that regard, macro-level economic stimulus packages need to be combined with catalytic climate finance for specific programs such as this one to ensure that this is achieved. If in 2017 those countries represented 1.2 percent of the total global GHG emissions, under a BAU scenario they would on average triple their emissions between 2017 and 2030. With international support, the seven countries are committed to contribute to GHG emissions reductions. As per their NDCs presented in *Figure 1*, Botswana, CAR, DRC, Kenya, Mali, Namibia and Uzbekistan committed to reduce, conditionally to external financing with the exception of Botswana, compared to their BAU scenario their emissions by 15 percent, 5 percent, 55 percent, 30 percent, 30 percent, 89 percent and 10 percent, respectively. However, the priority of countries is the immediate emergency response to the current Covid-19 crisis. The question of how the economic recovery is designed remains crucial in shaping the long-term pathways for emissions and determining whether the NDC targets can be achieved. If government do not roll out low carbon development strategies and policies in response to the coming economic crisis, emissions could rebound and even overshoot previously project levels by 2030, despite lower economic growth in the period to 2030.

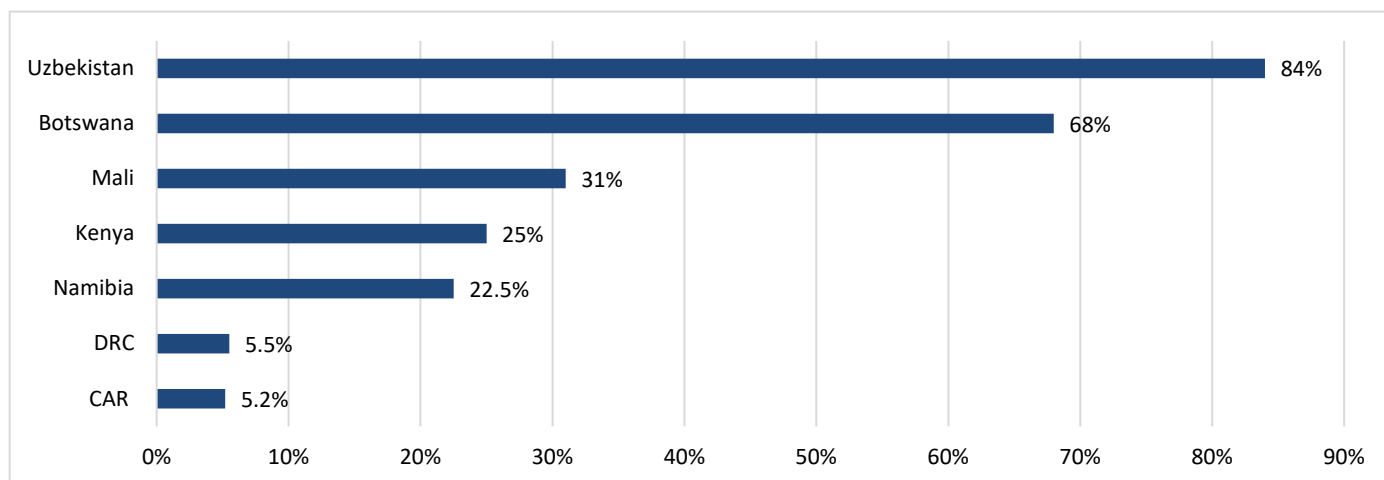
Figure 1: NDC target (conditional and unconditional – when applicable)



Sources: countries' NDC UNFCCC submissions

To reach their NDC targets, the seven countries will need to transform their energy sector. The energy sector represents in most of these countries the main contributor of emissions. The energy sector represents 84 percent of GHG emissions in Uzbekistan, 68 percent in Botswana and Mali 31 percent.

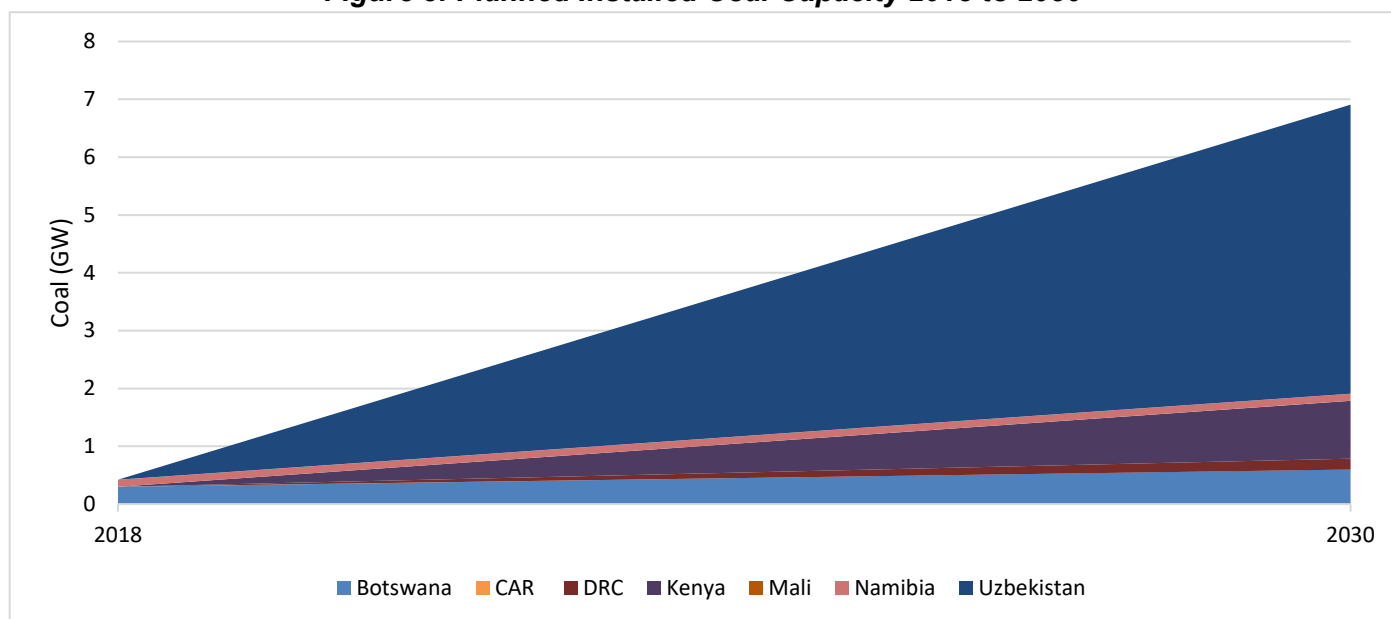
Figure 2: Energy Sector GHG Contribution per Country



Sources: countries' NDC submissions and various online sources

However, if the seven countries have ambitious GHG emissions reduction objectives as part of their NDCs, their current energy policies in place, the planned deployment of coal and the Covid-19 crisis may inhibit the NDC achievements. In the seven countries, total installed coal capacity is expected to increase from 0.5 GW to 7 GW from today's levels to 2030. As per their current plans by 2030, 8.5 GW of solar and wind generation will be built out of a total of 60 GW installed capacity – representing 14 percent. If representing these shares in GWh energy production, the solar and wind share would drop to 6-7 percent.

Figure 3: Planned Installed Coal Capacity 2018 to 2030



Sources: countries existing generation plans

It will be critical in this context to support policymakers to develop green stimulus interventions with targeted public investments in order to unlock large private investments for economic recovery.

According to the IEA and estimations by the World Bank, around 20 GW and 45 GW of RE (including large hydropower) by 2025 and 2030, respectively, will need to be deployed in those seven countries as part of their electricity mix to reach the 2C° targets and universal access. This represents around USD 15 billion in investments needed just for new solar and wind generation by 2025. To reach this objective, large amounts of private funding will have to be unlocked to complement the limited public sector financing available. Yet most developing countries still lack a pipeline of bankable solar and wind projects for consideration by the private sector. To develop one, countries must take a series of key steps to tackle critical risks perceived by the private sector such as grid constraints and procurement while also minimizing risks for the public sector.

The seven countries have been receiving support from other multilateral development banks (MDB) and bilateral banks to deploy RE in their countries outside of SRMI and of the scope of the present Funding Proposal – with a coordination at World Bank sector strategy level to ensure complementarity of programs, avoid overlaps and enhance synergies. More precisely:

- a. **Botswana:** The new World Bank-supported solar program is aligned with current support provided by Power Africa and the AfDB under a memorandum of understanding (MOU) on solar deployment.
- b. **CAR:** MDBs and donors are focusing their support in CAR – which is all grant – on access and generation: (i) the World Bank is financing an ongoing project consisting of a 25 MWp solar plant with 25 MWh battery storage – the PURACEL project, (ii) AfDB is currently financing the extension of Boali 2 hydropower system from 10 MW to 20 MW, plus the upgrade of the (only) transmission line between Boali and Bangui to 110 kV, (iii) AFD is working on refurbishing the existing 10 MW of Boali 2, (iv) the Chinese Government is supporting another solar project in Bangui (15 MWp), and eventually the construction of Boali 3 hydropower plant (5 MW) and (v) a Saudi fund is supporting investment in thermal generation for Bangui (10 MW). The new World Bank grant-funded project aims to transform the current 100 percent publicly owned energy sector by supporting the first steps towards mobilization of private investments and such, while supporting further deployment of disruptive technologies such as battery storage – CAR having the largest battery storage project being built in the African continent under the PURACEL project.
- c. **DRC:** The World Bank has been financing investments and technical assistance in DRC focusing on grid-connection, private investment mobilization for electrification, technical and financial support to the utility and interconnection to the neighboring countries. The World Bank program is aligned with the current portfolio of other MDBs such as AfDB, AFD, United Nations Development Program (UNDP), Japan International Cooperation Agency (JICA), the UK Department for International Development (DFID), USAID and the KfW. As the electrification agenda is critical to DRC while there are limited domestic financial resources, support for PPP RE mini-grids have been promoted under a DFID technical assistance financed by AfDB and GCF. The new USD 500 million DRC World Bank-supported project aims to leverage the lessons learnt from this PPP program and develop at a larger-scale fully privately financed mini-grids in some of the biggest regional capital cities. To unlock private financing for mini-grids, the project will propose tailor-made guarantees to the selected private investors to reduce the risk perceived.
- d. **Kenya:** The World Bank has been working on last mile electrification, generation (mostly geothermal) and utility strengthening/capacity building in Kenya as part of a large program funded by various donors. Since 2016, the World Bank Group, AFD and EIB are supporting solar and wind development with AFD financing a strategy on solar auction. The solar auction program has not yet been implemented due to limited appetite of the utility. However, the government has requested the World

Bank to support the modification of the solar auction program under its new RE program which will include least-cost generation plan, VRE integration and IPP procurement support. EIB financed IPPs in renewables and accessed GCF funding for its Global Energy Efficiency and RE Fund, and World Bank Group, through MIGA, is assessing providing support to RE IPP projects.

- e. **Mali:** The World Bank energy portfolio in Mali is focusing on access, sector strengthening and new generation. As part of its support to the West Africa Power Pool (WAPP), the World Bank is supporting the deployment of two regional solar with battery plants. These are being developed in coordination with other MDBs that are also supporting the utility and the energy sector, namely AfDB, the Islamic Development Bank (IsDB), the European Union, the West African Development Bank (*la Banque Ouest Africaine de Développement*, BOAD), AFD and KfW. These partners are preparing specific projects in the electricity sector in Mali and working in partnership on supporting the deployment of solar generation – with the World Bank leading the support to the VRE grid upgrades investment and the first deployment of competitive selection of IPPs. The ongoing OMVS 225 kV transmission line from Bamako to Manantali, Kayes to Senegal, and Kayes to Mauritania is funded by the AFD, IDA, and China Exim Bank. The IsDB is funding the mini-grids with productive uses in rural remote areas; the AFD is preparing a rural electrification program for some 60 localities; and the AfDB, AFD, and BOAD are co-financing the Guinea-Mali interconnection with the World Bank. GCF financing was accessed by BOAD for its climate finance facility to scale-up solar energy investments in francophone West Africa and its Mali solar rural electrification project. Clean Technology Fund (CTF) financing is being sought for the new investment project to support Mali in this transformative program with some of the largest solar and battery plants in the world (subject to CTF approvals).
- f. **Namibia:** The new World Bank-supported solar program is aligned with current support provided by Power Africa and the AfDB under a MOU on solar deployment. KfW and GIZ have also been supporting NamPower – the utility – in developing their least-cost generation plans and grid analysis as well as a battery storage pilot. Proparco provided debt financing for the 37 MWp PV. The Deutsche Bank accessed GCF funding for its universal green energy access programme. Together with the other MDBs and Development Finance Institutions (DFIs), the World Bank is coordinating its RE support to ensure complementarity of the different work streams.
- g. **Uzbekistan:** The World Bank Group has been very active in solar deployment and such with the support of ADB and EBRD. ADB financed a solar roadmap that was implemented with the support of the World Bank Group Scaling Solar Program with 1 GW of solar PV targeted. Asian Development Bank (ADB) is further supporting the deployment of solar and EBRD is supporting wind deployment in the country. Strong cooperation is ongoing to ensure that the tenders have similar bankable risk allocation. The new World Bank-supported VRE program in Uzbekistan will enable the integration of the VRE being deployed under the three MDB supported programs.

Outside of the scope of the present Funding Proposal, to facilitate the development of the Initiative's core objectives, a collaborative multi-stakeholder group has been established. The multi-stakeholder group comprises of AfDB, KfW, and EIB in addition to AFD, ISA and IRENA. The multi-stakeholder group is an informal high-level platform where stakeholders are expected to discuss SRMI's core principles during workshops sharing each stakeholder's vision to support the development of renewables in developing countries. A high-level collaboration with the Initiative stakeholders is ongoing (outside of the present Funding Proposal), including with AfDB's team working on their initiative Desert to Power and their new SEFA initiative on developing RE aligned with SRMI's principles, and AFD's work program on RE

deployment. Discussions are also on-going with the Energy Transition Coalition and in particular the Climate Investment Platform (CIP) to enhance cooperation and maximize synergies.

B.2. Theory of change (max. 1000 words, approximately 2 pages plus diagram)

The SRMI Facility aims at tackling the lack of sustainable and bankable pipeline of RE projects in developing countries to achieve low-emission development pathways. The SRMI theory of change is for the entire Programme and draws on the lessons learned from the successes and failures of national electricity policies and IPP selection processes in developing countries. Under the current context, it also aims to support the development of renewable energy in parallel with green stimulus packages for these countries to provide core support in the economic recovery post Covid-19. The SRMI Facility aims at supporting multiple countries in their energy transition starting with seven pilot countries and quickly after adding other countries via a complementary funding proposal presenting information on the new countries, such as Indonesia and Vietnam, once NOLs are received, for projects following the Facility's eligibility criteria presented in the present Funding Proposal.

The main barriers identified that need to be addressed to leverage private investments can be grouped into two broad categories: (i) risks occurring during the development phase, that is, prior to construction and operation; and (ii) those that arise once the project begins to operate. Both types of risk are integrated into the IPPs and lenders' cost of capital. These risks have been exacerbated under the current crisis, and comprehensive risk support will need to be provided to leverage private investments.

a. **Development risks** encompass:

- grid risk, including connection risks;
- land risk, including availability, permitting, and environmental and social (E&S) aspects;
- legal risk, including the applicable regulatory, dispute resolution, and judicial frameworks;
- procurement risk; and
- integrity risk.

b. **Operational risks** encompass:

- off-taker credit risk (including the off-taker's record of performance and timely payment) and risk of contract termination;
- the country's power sector risk (including sector financial sustainability risk, reform risk, regulatory risk, and delay in the government's construction work;
- market risk (including currency risk and interest rate risk;
- country and macroeconomic risks; and
- political risk (including risks of breach of contract; expropriation; transfer restriction and currency inconvertibility; and war and civil disturbance).

To tackle those risks, the SRMI Facility developed activities around four components – that are adapted to the needs of each country as part of the Facility, namely:

- a. **Component 1: Technical Assistance** – concentrates on: (i) comprehensive and integrated planning support to develop least-cost generation plans combined with VRE integration analysis and dispatch and planning capacity building, (ii) legal, financial and procurement aspects capacity building and transaction advisory support to implement competitive and transparent tendering of privately-financed RE project, and (iii) technical and environmental and social (E&S) studies to prepare solar and wind parks for future tenders;

- b. **Component 2: Public Investments** – concentrates on public investments for: (i) solar and/or wind park common/shared infrastructure, (ii) VRE integration grid upgrades (including publicly-owned battery storage) and enhancing grid resilience to climate change, and (iii) electrification to increase population's resilience;
- c. **Component 3: Risk Mitigation Instruments** – concentrates on providing risk mitigation instruments to private investors for grid-connected and off-grid renewable energy/storage projects; and
- d. **Component 4: Project Management** – concentrates on the management of the projects and coordination and implementation of their activities, in compliance with the Executing Entities (EEs)' and Program Implementing Entities (PIEs)' contractual obligations included or referred to in the legal agreements entered into with the Accredited Entities (AE).

The technical assistance component focusses on the country/macro level risks looking at (i) developing sustainable RE targets to reduce planning risks and provide evidence-based targets for low-emission pathways, (iii) reducing procurement risks through a robust and transparent selection of IPPs; and (iii) increasing government's internal capacity, and reducing developing risks for IPPs by providing land and permits through solar/wind park schemes.

The public investment component focusses on the grid and infrastructure level risks looking at (i) ensuring minimal curtailment and risk to the grid stability due to high penetration of VRE, (ii) increasing grid resilience to climate change impacts, finance the public infrastructure needed for the IPPs and (iii) providing access to electricity to vulnerable populations.

The risk mitigation instrument component focusses on the residual risks for private investors looking at ensuring a sustainable leveraging of private investments in grid-connected and off-grid RE projects.

Table 1: Components Financing per Project (indicative financing in million USD)²

Components	Total Financing Instrument	Botswana	CAR	DRC	Kenya	Mali	Namibia	Uzbekistan
Component 1: Technical Assistance								
Sub-Component 1.1 Generation plans with VRE integration and Capacity Building (technical)	Grants	0.5		1	1	1	1	0.5
Sub-Component 1.2 Transaction Advisory and Capacity Building (procurement, legal)		1.5	1	1.5	1	1	1	1.5
Sub-Component 1.3 RE Park Infrastructure studies		1.5	0.5	1	1	2.5	1.5	1.5
Component 1 indicative budget per country (all)	23	3.5	1.5	3.5	3	4.5	3.5	3.5
Component 1 indicative budget (GCF)	21.5	3.5		3.5	3	4.5	3.5	3.5
Component 2: Public Investments								
Sub-Component 2.1 Public investments in solar and wind park infrastructure	Loans (with the exception of CAR which is grant)	30	20			25	21	50
Sub-Component 2.2 Public investments for VRE integration grid upgrades and grid resilience		68	10		75	46	40	250
Sub-Component 2.3 Public investments for electrification (PPP mini-grids, grid extension)		49	60	500	125	62	35	
Component 2 indicative budget per country (all)	1466	147	90	500	200	133	96	300
Component 2 indicative budget (GCF)	206	30	30		45	38	20	43
Component 3: Risk Mitigation Instruments								
Sub-Component 3.1 Risk mitigation instrument for grid-connected RE	Reimbursable Grants			30				
Sub-Component 3.2 Risk mitigation instrument for mini-grids	Guarantees	10					10	
Component 3 indicative budget per country (all)	50	10		30			10	
Component 3 indicative budget (GCF)	50	10		30			10	
Component 4: Project Management (Grants)								
Component 4 indicative budget per country (all)	24.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5
Component 4 indicative budget (GCF)	2.5	0.5	0	0.5	0.5	0.5	0	0.5
Total indicative budget per country	1563.5	164	95	537	206.5	141	113	307

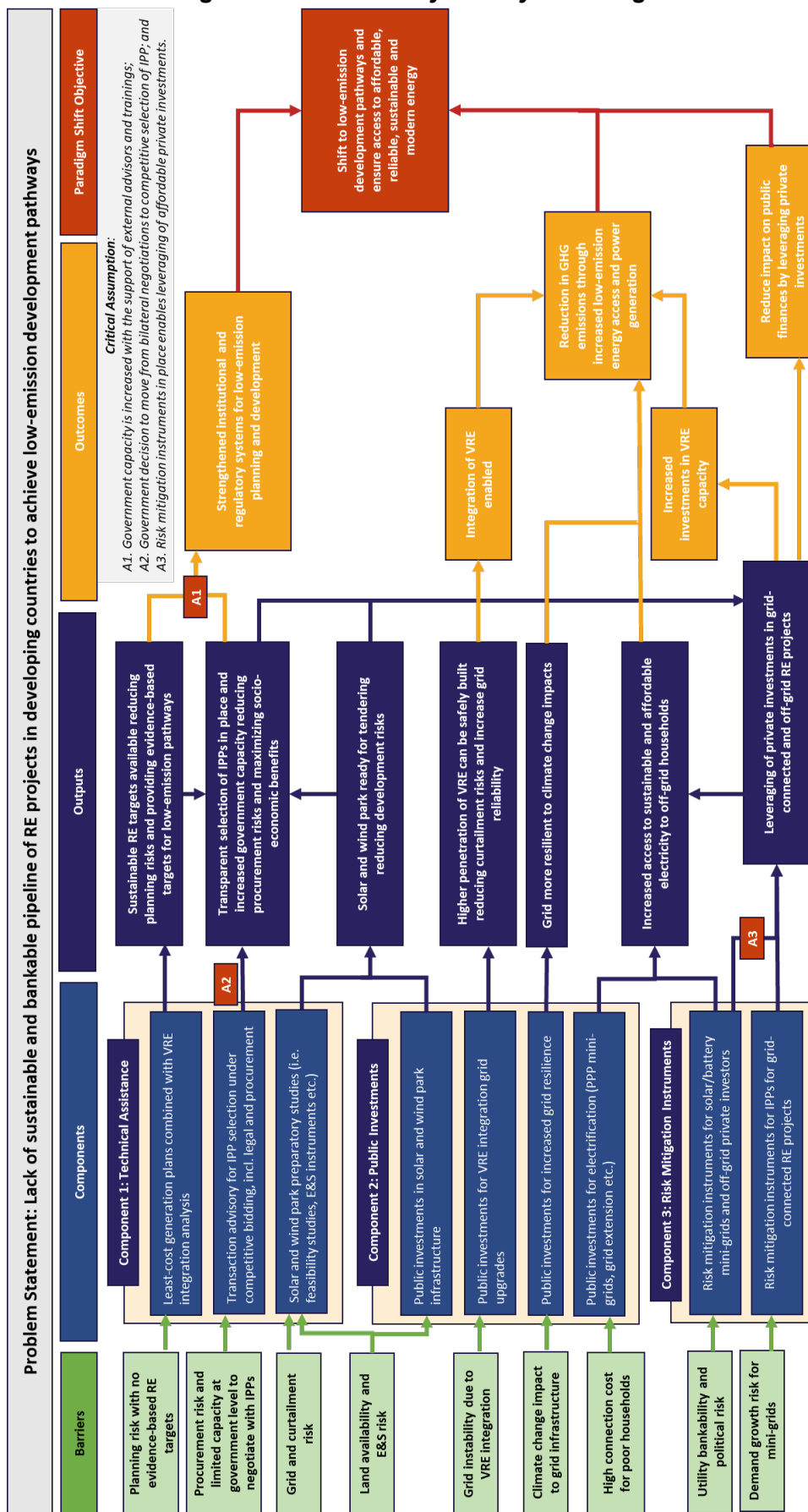
The main expected outcomes under the SRMI Facility are:

- a reduction in GHG emissions thanks to an increase in investments in VRE capacity through private investments in grid-connected and off-grid projects, and enabled VRE integration,
- the leveraging of private investments reducing the burden on public sector finances,
- improvement of grid resilience as well as resilience of local communities around the projects, and
- strengthening of institutional and regulatory capacity.

The SRMI Facility's objective is to support the eligible countries to shift to low-emission sustainable development pathways and increase access to affordable, reliable, sustainable and modern energy to its populations.

² All the financing presented in Table 1 are estimates based on current scope of the projects. These are indicative and subject to approval by the World Bank Board and agreements with the EE.

Figure 4: SRMI Facility Theory of Change



B.3. Programme description (max. 2000 words, approximately 4 pages)

■ Programme Overall Background

Solar and wind energy generation, as part of a green stimulus package post covid-19, could become game changers as they are already least-cost source of generation in most countries and are modular with shorter construction timeline than other power plants. Solar and wind energy generation can play a critical role from different perspectives: (i) climate change, (ii) energy security, (iii) energy access as it is a modular source of energy, (iv) responding to fast growing demand as they have short construction timeline, and (v) local development with industrial integration and community development (including from women empowerment's perspective). Moreover, solar and wind can help countries deal with their potential power shortages while ensuring they are not dependent on imported fuels.

However, large penetration of VRE is usually rapidly constrained due to grid technical limitations and lack of flexibility in the grids. The main challenges related to the integration of VRE into the grid are its variable nature, uncertain availability and that it can only produce power during the day for solar and when the wind is blowing. Three issues must be considered when deploying VRE into a grid: (i) the capacity for the generation mix to meet demand at any hour of the year, considering VRE variability; (ii) the economic optimum in terms of power system operating costs, after considering both cost reductions thanks to VRE and the investment costs required for deploying and integrating VRE; and (iii) the limitations of solar PV/wind capacity to maintain grid stability, due to its variability and its limited capacity to contribute to the balancing of demand and generation. The flexibility needed in the grid to enable a sustainable penetration of VRE can be constrained due to (i) technical limitations, i.e. can the grid have the dispatch and transmission capacity to integrate such variability, and (ii) commercial arrangements, i.e. even if there is in a given grid very flexible generators such as gas plants, they may not be able to provide flexibility to the grid as ancillary services may not be priced in their power purchase agreement (PPA) or they would have "take or pay" arrangements pushing the utility to dispatch them prior to considering new renewables.

Investment for grid modernization and upgrade will be required to ensure large penetration of VRE in the seven countries. Taking advantage of variable renewable generation requires significant expansion and modernization of electrical grids. Specific technologies and processes may be used to support the gradual transition of power systems into "VRE-friendly" grids that will significantly reduce integration costs in the long term. The penetration of VRE requires power system planning and grid management to adapt to the particular characteristics of VRE. It also requires better forecasting methods and stringent grid code requirements. In addition, this may provide the opportunity for the country to also look at its grid from a climate change resilience perspective and adapt it to future shocks.

If the technical aspects are critical, the deployment schemes selected to mobilize and select IPPs are also crucial. Since 2015, most countries that moved into competitive selection of IPPs directly benefited from the reduction in solar construction and equipment, and in cost of capital, through lower PPA prices compared to countries with FIT policies or bilaterally negotiated deals. As of the date of this Funding Proposal, Vietnam, Kenya and Mongolia have FIT prices of USD 9.35 cents, USD 12 cents and USD 15 cents per kWh, respectively. In comparison, in 2019, under competitive selection, Tunisia, Ethiopia, Zambia and Philippines announced PPA prices of USD 2.44, USD 2.56, USD 3.9 and USD 4.5 cents per kWh, respectively. The main difference between FIT and competitive bidding is that under FIT, Governments are fixing the price whereas under a competitive bidding process the private investors are.

Under competitive bidding, governments can access lower prices as IPPs will reflect in their proposed PPA price: (i) the capital expenditure (CAPEX) cost at the time of the bid – since 2010 CAPEX in wind and solar decreased rapidly, in particular solar module cost and it is difficult to assess its cost a few years ahead, (ii) high competition between solar IPPs leading to reduced equity returns expectations, (iii) bankable contractual arrangements with fair risk allocation between the private and public stakeholders that enable IPPs to attract long-term non-/limited recourse project finance that lower the cost of capital, and (iv) the scale of the market in the country and the region. Therefore, a bankable well-organized competitive selection can attract cost-competitive IPPs and ensure low cost of capital.

Based on the identified development challenges and in particular grid curtailment risks for VRE, two deployment schemes, namely substation-based competitive bidding and solar/wind park competitive bidding, are recommended. Both schemes aim to reduce the development risks perceived by IPPs, thus reducing the risk premium in the cost of capital and reduce the utility's cost of integrating solar generation into the grid. The main outcome expected is a reduced PPA tariff proposed by developers. More precisely:

- **Under the substation-based competitive bidding scheme**, the government sets a target of solar PV capacity to be procured at a certain time. It identifies substations/feeders with available MW capacity, and a certain MW capacity at each substation/feeder is opened for bidding to achieve the target. IPPs compete through a competitive bidding process to obtain the right to connect their projects to the identified substations/feeders based on a set of pre-determined selection criteria comprising technical, financial and safeguard aspects. IPPs are responsible for selecting the land, obtaining the investment license and have to follow the E&S requirements as required in the bidding documents. This scheme support countries better manage the volume of new solar PV/wind installation connected to the grid and develop RE in selected areas. It will enable the utility to optimize the use of existing transmission capacity, reducing the potential curtailment and/or additional costs to integrate the solar PV and wind projects. It can also proactively drive grid investments needed for new VRE generation.
- **The solar/wind park concept**, with pre-arranged comprehensive de-risking measures (including land, power evacuation and permits secured), can contribute to reach fully cost-effective electricity tariff. The government identifies project site(s), conducts land clearance, and constructs shared infrastructure for the solar/wind park that can range from the evacuation line to basic elements, such as the fence, roads, street lighting. The government conducts the E&S assessments as well as the geotechnical studies. Once the project is ready for competitive bidding, the bidding procedure begins and the winning IPP is responsible for the financing, construction, and operation of the solar/wind energy project. This scheme is critical when IPPs are identifying risk of curtailment and land ownership issues as paramount to the deployment of solar and wind in the country.

RE deployment can also be an opportunity for countries to maximize socio-economic benefits which is particularly important in the Covid-19 context. The lack of visibility on the local opportunities, the tight timeframe of the tenders, the existence of suppliers already tested, and the costs to assess the potential of the local industries do not encourage the bidders to explore what could be done competitively in a given country. Assessing the positioning of the local players on the solar/wind value chain and identifying the measures needed to improve their involvement in a competitive manner on such a value chain will increase their chances to benefit from the opportunities offered by the new RE projects. Factoring these measures in an RE roadmap and in the bidding processes launched can increase the participation of local players in a competitive manner, fostering the transfer of knowledge and the skills

development and allowing them to build a track record to the extent possible. In addition, expanding the RE roadmap to assess the needs of the local communities around the RE projects implemented and to design adequate programs to increase their resilience and enhance their livelihoods will ensure that they effectively benefit from the implementation of the projects. Sites with good solar/wind resources and large plots of lands are usually located in areas where the level of living is below the national average. The local communities there face numerous challenges hindering their development. Including tailor-made development programs in an integrated manner in the RE roadmap of a given country and operationalizing them through the RE projects deployed will ensure local communities benefit positively from them.

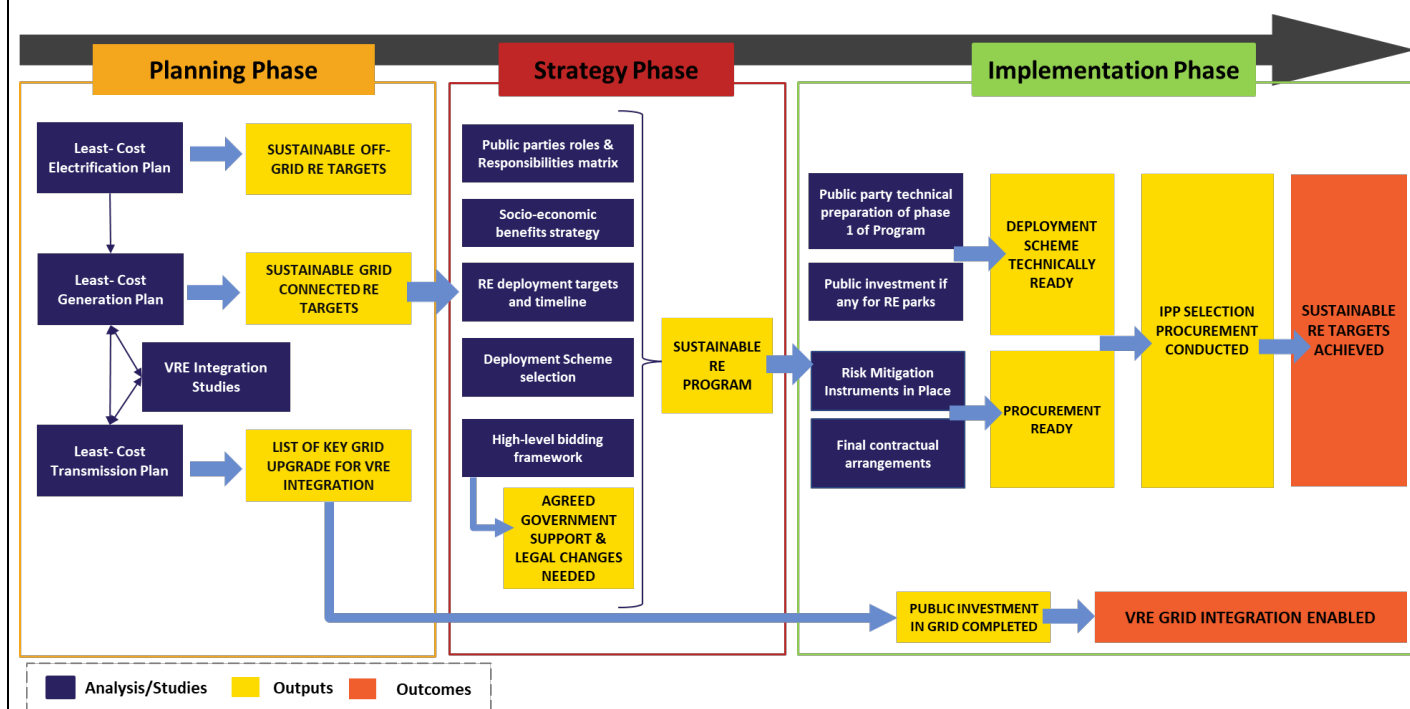
These programs aim to maximize the socio-economic benefits of the RE projects from the local industry as well as from the local community perspectives will pay a specific attention to women empowerment. Employment opportunities created by large-scale solar and wind infrastructure projects often benefit men more than women, resulting in a significant employment gap in the RE industry. Also, in the remote areas where large projects are developed, women, if employed, are more exposed to lower-paid, lower-quality employment in the informal sector. As a result, without a proactive policy to promote female employment throughout the RE project-cycle, men may benefit disproportionately from new employment opportunities across the RE value chain.

The World Bank SRMI aims at addressing in a comprehensive manner the different challenges that are limiting the uptake of sustainable privately financed RE and the associated socio-economic benefits. Launched in 2018 for the COP 24 under the leadership of the World Bank in partnership with the AFD, IRENA and ISA, SRMI aims to support countries in developing sustainable solar and wind programs designed to attract private investments and so reduce reliance on public sector financing, while maximizing the socio-economic benefits triggered by the projects deployed. Its unique and integrated approach offers development and climate financing for (i) technical assistance to help countries develop evidence-based RE targets, implement a sustainable renewable energy program, and set up and maintain transparent and competitive procurement processes with transaction advisors; (ii) critical public investments to enable integration of VRE, finance solar/wind park infrastructures, and increase access to electricity; and (iii) risk mitigation instruments to cover residual risks perceived by private investors. Its framework is supporting the creation of jobs, the development of skills as well as increasing the resilience and enhancing the livelihoods of the local communities around the projects.

SRMI aims to support countries to:

- develop sustainable RE programs to leverage private investment at scale for grid-connected and off-grid RE projects,
- implement the country's RE programs through transparent and robust competitive tenders,
- improve their grid capacity to integrate VRE and resilience to climate change,
- leverage private investments for RE grid-connected and off-grid projects,
- build RE capacity,
- avoid GHG emissions, and
- maximize the socio-economic benefits triggered by the RE projects developed.

Figure 5: SRMI Methodology



Country Context

The SRMI approach aims to support countries in a comprehensive multi-dimensional risk mitigation approach, yet in a flexible manner adapting to their current situation and constraints. Therefore, it is critical to understand the different constraints and starting points of each of the countries currently identified to benefit from the Facility:

- **Botswana's** power sector currently relies on high-carbon electricity, largely produced from coal at the Morupule power station using the country's large coal reserves (around 200 billion tons). To meet domestic demand, 30 percent of electricity is imported from South Africa (92 percent coal) and the SAPP market (74.5 percent coal) contributing to coal-fired emissions and unnecessary fiscal outflow as tariffs do not cover peak power prices. Only sixty percent of national households in Botswana have access to electricity, and the government wants to reach universal access by 2030. Under a BAU scenario, Botswana could see increased output of coal-fired power generation, and annual emissions of greenhouse gases will be 2.5 times their 2012 levels by 2030. Yet, Botswana enjoys one of the highest irradiation densities in the world, with over 3,200 hours of sunshine per year and an average global irradiation throughout the country. This resource, while largely underexploited today could place Botswana as a leader for developing solar. Botswana also benefits from a good wind resource, complementary with the solar resource, with a potential currently estimated between 300 MW and 500 MW. No solar/wind projects are built to date in the country.
- **CAR's** electricity sector is characterized by inadequate infrastructure, a weak policy and regulatory framework and a utility that is struggling to recover costs and thus maintain and expand services. The overall rate of access is 8 percent in the country. Total installed capacity is 28 MW, of which 23 MW is available, whereas the current load is 45 MW, and the projected unmet demand is estimated at around 60 MW. The majority of the generation is hydropower and diesel plants. Therefore, to improve reliability in the power system, support stability, and promote

peacebuilding in a context of fragility, the country is in dire need of sizeable and urgent increases in its electricity generation capacity. Solar with battery storage is being sought as a solution with a World Bank financed utility-owned solar plant under construction and a deployment of solar/battery mini-grid, and off-grid solutions. The country does not have yet any privately-owned generation but has a provision for IPPs in its new Electricity Law and started working on the decrees for the implementation of the law.

- **DRC** has an access rate of around 9 percent. The well-distributed potential of both hydro and solar resources remains mainly untapped, as power sector development has focused on a few large hydropower plants over the last decades, leaving most of the country's population unelectrified. SNEL, the national state-owned utility, has a critical role to play in access, in particular in Kinshasa, but is caught in a vicious cycle of mounting commercial losses, deteriorating assets and mounting debt that hinder its ability to drive electrification outside of Kinshasa. Therefore, it will be critical for the future of the electrification in DRC to develop large solar/battery mini-grids in off-grid zones by leveraging private investments. To enable private investments, the government must improve its institutional and regulatory framework.
- After decades of shortages, **Kenya** now has enough generation capacity to meet its demand, and supply is predominantly from renewable energy sources. The current installed generation capacity is 2,819 MW with a peak demand of 1,912 MW, resulting in a peak reserve margin of over 30 percent. Renewable energy (geothermal and hydro) account for 65 percent of total installed capacity and almost 90 percent of total electricity generation. Geothermal energy contributes more than 45 percent of total energy generated, making Kenya one of the global leaders in use of this firm, renewable resource. Kenya has a FIT since 2008, and revised in 2012, for wind, biomass, small-hydro, geothermal, biogas and solar with a standardized PPA. However, little solar and wind generation were built under the FIT, with only 65 MW and 335 MW, respectively.
- Despite significant progress over the last decade, **Mali's** access to modern energy services remains low and overall unreliable. Mali relies heavily on expensive thermal power generation (heavy fuel oil (HFO) and diesel) and have an access rate of around 39 percent. The development of interconnections with its neighboring countries is enabling the country to reduce its cost of electricity, while improving its reliability. However, to ensure energy security, the country have been bilaterally negotiating with IPPs for new solar projects. No solar/wind projects are built to date as prices negotiated are well above the regional standards (with prices well above USD 0.10 per kWh). In addition, Mali's grid is under development with key backbones under construction and therefore, will face issues with integrating large amounts of VRE without any critical investments put in place first.
- **Namibia** imports 73 percent of its electricity needs and is particularly reliant on South Africa's imports. High imports also lead to high foreign exchange, high electricity costs which place a burden on the balance of payments and budget, and vulnerable electricity supply. The government has taken an important policy decision to increase domestic supply and reduce electricity import-dependency through the procurement of public and private investments in electricity generation and transmission. Namibia aims to achieve 80 percent self-sufficiency through domestic electricity production by 2022. Namibia is one of the best countries in the world for CSP, PV and wind production. Namibia has an experience in FIT for very small VRE projects

and a limited experience in competitive bidding. Namibia has an installed 120 MW of PV capacity and 5 MW of wind.

- **Uzbekistan's** energy system is characterized by high losses and low reliability of supply due in part to the rapidly aging supply infrastructure that was mostly constructed during the Soviet era. The energy sector is highly dependent on natural gas, which is a major source of commodity exports but getting depleted and making the system vulnerable, thereby prompting the country to seek sustainable energy transition pathways. Natural gas accounts for 86 percent of the total primary energy supply. Diversification of Uzbekistan's electricity generation is also limited, as more than 80 percent of the total installed capacity of 12.5 GW in Uzbekistan is natural gas based, followed by hydropower (16 percent) and coal (11 percent). The least-cost power generation expansion plan for Uzbekistan provides a path to solar and wind transition with solar and wind accounting for around 30 percent of the total capacity mix (7.7 GW). Uzbekistan has already a very successful tendering system in place with the support of the World Bank Group Scaling Solar that was implemented and reached prices of US\$ 3.9 cents per kWh. However, to be able to reach their ambitious VRE targets, Uzbekistan needs to tackle the limitation on its grid availability/flexibility which is expected once a higher penetration of VRE will be reached.

■ Programme's Objective & Components

The SRMI Facility's objective is to support the eligible countries to shift to low-emission sustainable development pathways and increase access to affordable, reliable, sustainable and modern energy to its populations. The SRMI Facility is developed around four components that are complementary. Depending on the country specificities such as access rate, available plans and grid flexibility, sub-components may not all be required for each project. These components are based on the identified risk mitigant that will enable to reduce the cost of RE deployment, contributing to low-emission development pathways. At individual project level, the Facility development objectives will be adjusted to reflect what is specifically done under each project and which objective and impacts the activities in each project are particularly designed to focus on.

The Facility and its Components have been developed with enough flexibility to be able to add countries to the Facility once (and if) approved by GCF Board under subsequent phases. The design of the Facility, its components as well as its instruments are applicable to all countries that aim to apply the SRMI approach to sustainably move forward with their energy transition.

Component 1: Technical Assistance

Grant amounts will be allocated by the EE (Host Country) to the financing of the following types of activities to be carried out by their Ministry or state utility responsible for energy, on an as country-needed basis:

Sub-Component 1.1: Generation plans with VRE integration and Capacity Building (technical)

- Preparation of a new national (or adaptation of an existing) least-cost power generation plan, which includes reiterative VRE integration analyses based on technical and commercial constraints, an evaluation of the dispatch capabilities and the limits of grid integration and determines a cost-optimized electricity mix that can meet demand at any time;
- Preparation of a new national (or adaptation of an existing) least-cost transmission/distribution plan, which includes capacity expansion models simulating generation and transmission capacity

investment, given assumptions about future electricity demand, fuel prices, technology cost and performance, policy and regulations, and providing informed solar targets and indication of where best to locate future projects as well as a list of key investments needed to improve grid VRE integration capacity; and

- Provision of institutional capacity building support to government institutions and/or utilities/dispatching entities (as identified and designated by the EE) for energy planning and dispatch (including using grid-connected battery storage).

Sub-Component 1.2: Transaction Advisory and Capacity Building (procurement, legal)

- Provision of technical assistance (including legal, procurement and transaction advisory services) and implementation of capacity building activities to support: (i) the assessment of the existing legal framework and the development of a legal framework for IPPs and a competitive bidding scheme that is conducive to private sector investment; (ii) the design of energy projects (including key terms of the PPA and the government support agreements) with fair risk allocation among stakeholders and contractual arrangements acceptable to potential investors; (iii) the competitive selection of IPPs and mini-grid developers; and (iv) the negotiations of the final contractual arrangement with the selected investors and closing.. If the relevant Project includes Component 3, the development, structuring and preparatory work for Sub-component 3.1 and Sub-component 3.2 will be expected to be financed and carried out under this Sub-component 1.2. Under such technical assistance, a specific attention will be brought to inform clients on how to leverage an effective and efficient competitive bidding process (including in terms of evaluation formula to select the IPPs) to maximize the socio-economic benefits, building on the results of the associated studies, lessons learnt and best practices.

Sub-Component 1.3: Solar and wind park preparatory studies

- Preparation of preparatory and feasibility studies and environmental and social risk mitigation instruments, for the development of solar PV, solar PV with battery storage, CSP, onshore wind and offshore wind parks in support of investment de-risking and investor (IPPs and mini-grids developers) selection process. Preparatory and feasibility studies may include, as needed, geospatial analyses of the land around the substation, topography and geotechnical analyses, environmental and social analyses, site-specific grid interconnection studies, solar irradiation/wind measurement analyses and other studies as needed.

Component 2: Public Investment

Loan amounts (or grant for grant only countries) will be allocated by the EE (Host Country) to the financing of the following types of activities to be carried out by their Ministry or state utility responsible for energy, on an as country-needed basis:

Sub-Component 2.1: Public investments in solar and wind park infrastructure.

- Carrying out infrastructure investments in support of the development of solar and wind parks, such as acquisition of land, construction of transmission lines and fencing of such infrastructure for solar and wind parks.

Sub-Component 2.2: Public investments for VRE integration grid upgrades and grid resilience

- Investment for grid modernization and upgrade to improve grid's flexibility to integrate larger amounts of VRE, limit curtailment or grid instability and adapt to climate change related shocks,

including power system planning, grid management, strengthening of forecasting methods and grid code requirements. Grid reinforcements that support VRE integration may include (i) addition or replacement of lines and transformers for grid extension and capacity enhancements (both for answering growing demand and for integrating VRE power), (ii) equipment for smoothing the voltage and frequency issues, such as capacitor banks, battery storage and other reactive power compensators, together with flexible alternating current transmission systems, (iii) equipment for faster and more efficient grid operation, such as monitoring systems, demand and production forecasting systems, and automats for controlling generation units and grid operations through automatic generation control; and (iv) other innovative technologies whose commercial viability may not have been established yet such as load shifting battery storage.

Sub-Component 2.3: Public investments for electrification

- Investments as needed to support increased electrification, in partnership with the private sector when feasible, including through solar home systems, mini-grids and grid connection, as best appropriate to the country context, with a particular focus on women.

Component 3: Risk Mitigation Instruments

Provision of financial instruments (in the form of guarantees or reimbursable grants whose reimbursement is conditional and modular based on verified performance, or other instruments derived from their financial engineering) to reduce the following risks of investors or their commercial financiers and enhance financial sustainability, in the following areas:

Sub-Component 3.1: Risk mitigation instruments for solar/battery mini-grids and/or off-grid private investors

- Provision of reimbursable grant to mitigate the critical risks related to private capital investment in the off-grid space (mini grids and/or SHS – for which Multi-Tier Framework (MTF) data will be used when available), such as for instance demand off-take risk affecting the investors for the first few years of operation and consequently financial risks affecting pricing and tenor of financing available to the investors.

Sub-Component 3.2: Risk mitigation instruments for grid-connected RE projects

- Provision of guarantees to mitigate the critical risks related to private capital investment and provision of credit enhancement to private sector investors (and financiers) and electricity off-takers as needed to leverage private sector investments in grid-connected RE projects. If GCF so approves, these risk mitigation instruments may be provided without member country counter-guarantee, meaning that any amounts paid under the risk mitigation instrument following the occurrence of a covered risk would not be reimbursed to the AE or to GCF.

Component 4: Project Management

Management of the projects and coordination and implementation of their activities, in compliance with the EEs' and PIEs' contractual obligations included or referred to in their subsidiary agreements entered into with the AE.

▪ **Programme's Project Eligibility Criteria**

Each of the Projects which will be included in the Program – under the present Funding Proposal and any subsequent phases - will have to meet the following eligibility criteria that are aligned with and organized along the GCF Investment Criteria, prior to being approved by the Accredited Entity's approval body and being able to access the Facility financing:

- **Impact potential:**
- The project will directly support: (i) the financing and the integration of solar and wind energy representing a minimum of 15 percent of future total installed capacity by 2025 in the given country (or grid), or (ii) the financing and the integration of a minimum of 60 percent of future solar and wind installed capacity by 2025 in the given country (or grid), or (iii) the financing of a minimum of 1 GW of solar and wind in a given Host Country (*excluding* projects that are not requesting financing for Sub-component 2.1 and 2.2).
- The project will support the country in its effort to leverage private investments in a competitive and efficient manner. This criterion's applicability is dependent on the current security and macro-fiscal constraints that the country may have. However, even in Fragile, Conflict-affected Situations (FCS) countries such as CAR, DRC and Mali, it is critical to develop a path to private sector leveraging to reduce the impact of large-scale infrastructure financing onto the government's budget.
- The project will not directly finance nor support non-RE sources and will instead support VRE deployment by financing (i) investment for grid modernization and upgrade to improve grid's flexibility to integrate larger amounts of VRE (using technology such as new SCADA systems, STATCOM, capacitor banks, substation upgrades with transformer, battery storage, transmission line upgrades/increased voltage, and new transmission lines), limit curtailment or grid instability (using technology such as battery storage and other climate friendly technologies - and other technical solutions which may be developed after the date of this FP, whose reliability and efficiency, based on the technical data available at the time, have been considered sufficient by the AE at the time of its internal approval of the project) and adapt to climate change related shocks (technology such as lines and substation following climate change resilience guidelines) (ii) and/or solar/wind park infrastructure (such as land, substations, new transmission line connecting the park to the grid, fencing, security system and all infrastructure needed to complete the park as the case may be), and/or (iii) renewable energy-based electrification (including solar home systems and mini-grids or any climate friendly technologies).
- If the project includes activities under Sub-component 3.1 or Sub-component 3.2, it will cover risks related to long-term mobilization of private capital selected in a competitive manner, and in RE investments in utility-scale or mini-grids.
- The estimated installed capacity (MW) as a result of infrastructure financed under the Facility shall not deviate more than 15 percent less than the initially planned installed capacity for the whole Facility as indicated in Table 2 below.
- The estimated mitigation impact (tCO₂eq) as a result of infrastructure financed under the Facility shall not deviate more than 15 percent less than the initially planned mitigation impact for the whole Facility as indicated in Annex 3 of the Funding Proposal.
- The project will support solar and/or wind (with or without storage) technologies.
- **Paradigm shift potential:**

- The project will focus on increasing RE deployment in the country (grid-connecting and/or off-grid) in a transformative manner – from different angles such as legal, procurement, technical, economic and social development etc., depending on the starting point of the country.
- The project will include a capacity building plan to be financed under the project to support the long-term impact of the change promoted under the project.
- **Sustainable development potential:**
 - The project will, to the extent possible, support the development of an industrial integration plan to support local industries, skills development and long-term jobs' creation.
 - The project will, to the extent possible, support the development of a local development action plan to increase the resilience and enhance the livelihoods of the local communities.
- **Country ownership:**
 - The relevant Host Country will demonstrate a strong willingness to develop a comprehensive and streamlined strategy to deploy and integrate VRE and leverage in a sustainable manner private investment.
 - The VRE targets and revised policy supported by the project will be aligned with the NDC/low-emission development plans as reflected in the no-objection letter issued by the Host Country.
- **Recipients' needs:**
 - The Host Country will be vulnerable to climate change and in need to fundamentally adapt its energy systems by harnessing its domestic renewable resources to mitigate climate change.
- **Efficiency and effectiveness:**
 - The project will aim at substantially reducing GHG emissions in the given country compared to BAU in the energy sector of the Host Country(ies) and support the achievement of the Host Country(ies)' NDC.
 - The project's expected ratio of total USD per carbon dioxide invested³ is maximum 45, and per GCF USD invested – the GCF Proceeds – is maximum 10 for each project *excluding projects that are not requesting financing for Sub-component 2.1 and 2.2, and full grant projects*
 - The SRMI Facility budget (GCF Proceeds) allocated per Host Country shall not deviate more than 15% percent from the initial budget allocated per Host Country as indicated in Table 2 below.
 - The project's financing will be structured with a minimum co-financing ratio GCF to co-financing of 3 (one to two co-financing) in each country.
 - The project will have been prepared in compliance with the Accredited Entity's applicable policies and procedures and the relevant provisions of the AMA and the FAA.
 - The project will be approved by the World Bank within 3 years after the Facility's GCF Board approval.
- **Gender:**

³ Defined as cost per ton CO₂ equivalent, calculated as total investment cost divided by expected lifetime emissions reductions.

- The project will contribute to closing gender gaps in the energy subsector by (i) identifying gaps relevant to the four pillars of the World Bank Group (WBG) Gender Strategy⁴, and (ii) aiming to address these gaps through specific actions supported by the project.

The projects presented in this Facility are aligned with the eligibility criteria proposed, as illustrated in the feasibility study annexed to this Funding Proposal. The World Bank will work with each Executing Entity covered under this Facility to finalize the scope and co-financing of these projects, taking into account (i) the needs of the clients as they may evolve, (ii) the results of the studies at project's level and (iii) the national dialog involving all relevant stakeholders. A multi-disciplinary World Bank team will closely coordinate with the operational teams to ensure that the preparatory work carried-out at the project's level is adequate for the projects to meet all the eligibility criteria as agreed under the Facility with the GCF.

▪ Identified Project's Scope and Financing

Subject to IBRD/IDA/ESMAP corporate approval as appropriate in due course, the projects currently considered for inclusion in the Programme are expected to be financed through GCF financing and a co-financing from IBRD/IDA in the form of loans and IDA and ESMAP in the form of non-reimbursable grants. Co-financing does not include any other sources of financing for separate (even if coordinated/complementary) projects/activities that fall outside the scope of this Funding Proposal nor private investments that are only leveraged. CTF financing may be sought for Mali – however, this financing is not yet approved. Given the limited CTF resources expected to be available, the contemplated CTF financing is expected to substitute a very limited part of the World Bank resources if the CTF resources are eventually approved. The CTF portion is currently included in the WB financing described in the present Funding Proposal and therefore there is no financing gap in what is presented in the Funding Proposal. For the avoidance of doubt, in the event that the CTF resources are approved, CTF resources will replace IBRD/IDA/ESMAP resources. In the event that the CTF resources are not approved, IBRD/IDA/ESMAP resources will continue to be used so there will not be any financing gap. The World Bank targets to submit a request for CTF financing for the Mali project upon opening of the new CTF window supporting VRE integration. However, as this window is expected to be operational by late Q3 2021 (as the operationalization of the associated modalities is still on-going), it is likely that the Mali project will be submitted to the CTF Trust Fund Committee for approval post the GCF Board targeted for the SRMI Facility (B28). Meanwhile, these projects are included in the CTF pipeline and the World Bank will be monitoring every opportunity to advance CTF financing to these projects.

Table 2: Indicative Projects Scope and Financing Needs (in million USD)⁵

Country	Scope of project	Comp. 1	Comp. 2	Comp. 3	Comp. 4	Total cost
Botswana	a. park shared infrastructure for 200 MW CSP and 100 MW wind,	3.5 (GCF: 3.5)	147 (GCF:30)	10 (GCF: 10)	3.5 (GCF: 0.5)	164 (GCF 44)

⁴ The four pillars: (i) improving human endowments, (ii) removing constraints for more and better jobs, (iii) finding solutions that help increase women's ownership of and control over assets, and (iv) enhancing women's voice and agency and engaging men and boys.

⁵ All the financing presented in Table 2 are estimates based on current scope of the projects. These are indicative and subject to approval by the World Bank appropriate bodies and agreements with the EEs. Component 1 financing instrument is grants, Component 2 financing instrument is loans except for CAR which will be full grant, and Component 3 is a mix between reimbursable grants for DRC mini-grids and the others will be GCF guarantees. For further information on the split between sub-components please see Table 1.

	b. VRE integration and grid upgrades including 50 MWh of storage, and c. electrification for estimated 600,000 people; and d. risk mitigation instrument covering liquidity/termination for private investors competitively selected for CSP and/or wind projects developed under SRMI.					
CAR	a. solar PV (15 MWp) with battery storage (15 MWh) and VRE integration upgrades, b. mini-grid (20 MW PV with 20 MWh storage) and off-grid RE solutions for provincial cities and rural areas to electrify estimated 250,000 people, and c. support energy sector and utility's financial performance to attract private sector participation in renewable energy generation.	1.5	90 (IDA grant) (GCF: 30)	-	3.5	95 (GCF: 30)
DRC	a. Public-private-partnership (PPP) 100 MW PV and 140 MWh storage mini-grids to electrify estimated 1.5m people, b. grid infrastructure and resilience, c. risk mitigation instrument to cover demand risks for private investors for mini-grids.	3.5 (GCF: 3.5)	500	30 (GCF: 30)	3.5 (GCF: 0.5)	536.5 (GCF: 34)
Kenya	a. park shared infrastructure for 300 MW grid-connected solar, b. VRE integration including 150 MWh of storage, enabling integration of 250 MW solar and wind, and c. electrification for estimated 420,000 people.	3 (GCF: 3)	200 (GCF: 45)	-	3.5 (GCF: 0.5)	206.5 (GCF: 48.5)
Mali	a. park shared infrastructure for 300 MW grid-connected solar with 300 MWh storage, b. VRE integration incl. 50 MWh storage enabling integration of 150 MW PV, and c. electrification for estimated 400,000 people.	4.5 (GCF: 4.5)	133 (GCF: 38)	-	3.5 (GCF: 0.5)	141 (GCF: 43)
Namibia	a. park shared infrastructure for 200 MW CSP and 200 MW wind, b. VRE integration incl. 200 MWh storage, c. electrification for estimated 1m people, and	3.5 (GCF: 3.5)	96 (GCF: 20)	10 (GCF: 10)	3.5 (GCF: 0)	113 (GCF: 33.5)

	d. risk mitigation instrument covering liquidity/termination for private investors competitively selected for CSP and/or wind projects developed under SRMI.					
Uzbekistan	a. park shared infrastructure for 1 GW solar PV, b. VRE integration including 50 MWh enabling integration of 1.5 GW VRE, c. grid upgrade and deployment.	3.5 (GCF: 3.5)	300 (GCF: 43)	-	3.5 (GCF: 0.5)	307 (GCF: 47)
		23 (GCF:21.5)	1,466 (GCF: 206)	50 (GCF: 50)	24.5 (GCF: 2.5)	1,563.5 (GCF:280)

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

The following section describes the overall Facility management, the management process to ensure the most transformational projects are selected, and to deploy the legal arrangements between GCF, World Bank and the countries.

1. SRMI Facility as a GCF Program

The SRMI Facility is a program to support a series of projects, that currently proposes to support seven projects subject to the results of their appraisal but possibly more. All projects which will ultimately be supported by the SRMI Facility will have the common purpose to scale-up RE in a systematic and integrated manner. This Programme aims to enhance coordination and strategic alignment of the projects in a manner that can increase the prospects of transformational change, provide operational efficiencies and accelerate implementation of the SRMI methodology across countries and regions. The Facility is being developed as a Programme to ensure enough flexibility to include countries under subsequent phases, such as Indonesia and Vietnam when the NOLs are received and upon confirmation of them meeting the eligibility criteria.

This Funding Proposal includes indicative information for each of the currently proposed seven projects. Under the programmatic approach, the GCF would rely on the AE's due diligence on the proposed projects included in the Program. Subject to the outcome of such due diligence, the World Bank will approve funding for such projects that meet the eligibility criteria as presented in Section B3.

The EE are responsible for ensuring the implementation of the projects and the fulfilment of their development objective. The AE will provide implementation support to the EE to the extent permitted under its policies.

2. Implementation Arrangements

As part of its role as GCF AE and co-financier of the projects to be supported under the Facility, the World Bank will assess and confirm the eligibility of each such Project to be included in the Facility and benefit from the Bank's and GCF Financing. In order to do so, for each project proposed to be included in the Facility, the World Bank will comprise energy and climate finance experts to manage and be responsible for the Facility's overall objectives and ensure that the projects selected meet all the eligibility criteria (and their limitations) as presented in Section B3.

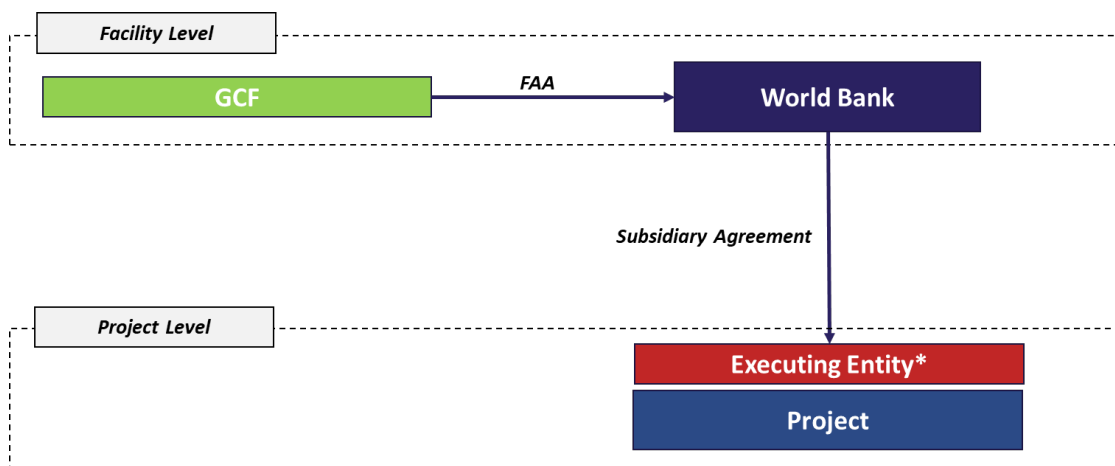
Following the GCF Board approval, the World Bank and the GCF will, based on the Accreditation Master Agreement (AMA), enter into one Funded Activity Agreement – FAA for the provision of the funds under all GCF instruments supporting the Program. The World Bank will be solely responsible for the management and administration of GCF resources and will carry out such management and administration in accordance with its policies, procedures and practices, and the relevant provisions of the FAA and AMA. The World Bank will apply its own fiduciary principles and standards relating to any integrity checks, anti-corruption, countering of financing of terrorism, fraud, financial sanctions, embargoes and anti-money laundering.

The World Bank will inform the GCF Secretariat about projects selected in accordance with the Eligibility Criteria as part of the periodic reporting outlined in Section E7. During the implementation of the Facility, the World Bank will be responsible for providing the necessary governance, oversight and quality assurance in accordance with its policies, procedures and any specific requirements in the AMA and FAA.

The World Bank will use its operation systems and established processes to review individual project proposals that would meet the eligibility criteria as presented in the present Funding Proposal and approve allocations. It would also facilitate the reporting in an integrated way to the GCF on the SRMI Facility's projects, as well as the learning across projects. Following approval of Projects, the World Bank will sign Subsidiary Agreements with the EEs and other agreements as applicable with any loan/grant PIEs and guarantee obligors and beneficiaries. These transactions will make available World Bank finance as well as GCF finance for investments in line with the provisions of the FAA.

There is no capacity assessment of the countries presently in the Facility, as those will be conducted during the appraisal stage of each project. When the projects go through the appraisal stage, some of the capacity assessment information will be summarized in the projects' appraisal documents and will be publicly disclosed. As part of the World Bank's due diligence in accordance with its own policies and procedures, WB ensures that adequate fiduciary (procurement and financial management [FM]), safeguards, legal and M&E frameworks are in place: (i) the different experts within the World Bank are assessing the client's capacity, (ii) they provide recommendations in the event where the capacity is not sufficient and (iii) an effective project management unit (PMU) with the core experts is usually part of the legal covenants for EEs to meet.

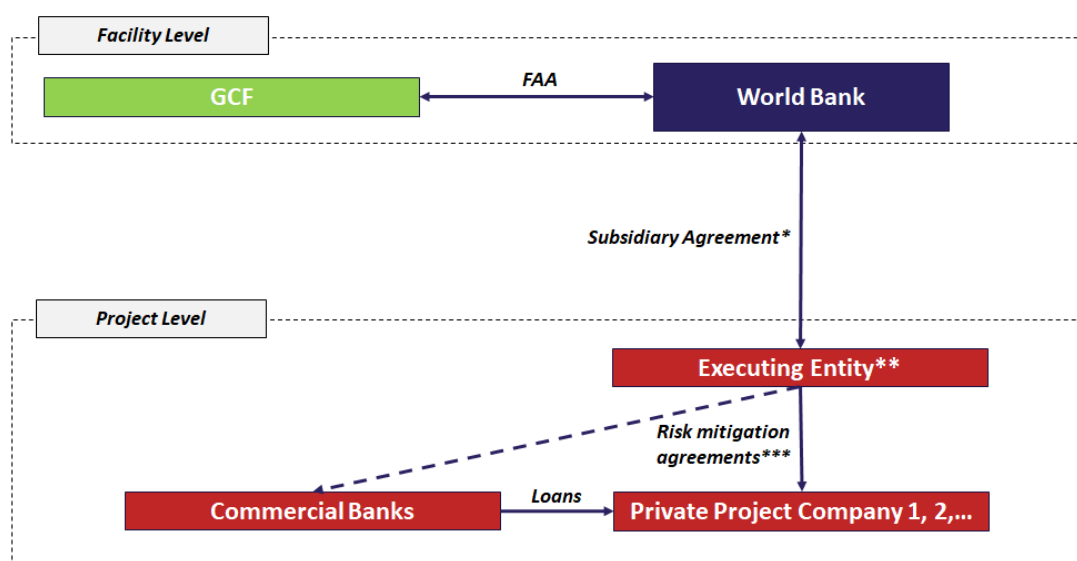
Figure 7: Indicative Implementation Arrangements for Component 1 and 2



*If a project has a Project Implementing Entity (PIE), there may be additional arrangements including a project agreement between the World Bank and such PIE to provide on-lending/on-granting.

With regards to risk mitigation instruments, depending on the final arrangements that will be developed as part of the preparation and appraisal of each project, figure 8 presents the links between the country, commercial banks and private developers for Component 3.1 and figure 9 for Component 3.2.

Figure 8: Indicative Implementation Arrangements for Component 3.1



* Including commitment to reimburse the grant remaining at the end of the implementation period.

**If a project has a Project Implementing Entity (PIE), there may be additional arrangements including a project agreement between the World Bank and such PIE to provide on-granting.

*** Payments structured to mitigate key risks, such as early demand risk.

Overview of the indicative mechanism contemplated for the reimbursable grants, subject to future due diligence, market sounding, and World Bank management approval:

- a. **Eligible sub-projects:** RE private sector projects selected by the Executing Entity, with the technical assistance of the World Bank to ensure that they meet both the eligibility criteria as well as the Bank's mandatory requirements (including safeguards and fiduciary) for sub-projects, as stated in the GCF financing agreement between the Accredited Entity and the Executing Entity
- b. **Executing Entity(ies):** a Host Country, expected to be the Democratic Republic of Congo, or a state-owned company mandated by the Host Country
- c. **Beneficiaries:** private sector developers preparing a public private partnership for the development of mini-grid sub-projects with an agreed business plan
- d. **Covered Risk:** mitigation of the critical risks related to private capital investment in the off-grid space, mainly demand off-take risk affecting the investors for the first few years of operation and consequently financial risks affecting pricing and tenor of financing available to the investors (to support viability during the initial years of the sub-projects)
- e. **Financial terms:**
 - Disbursement to the project company established by the private developers for the implementation of its sub-project, during a pre agreed number of years at the beginning of the operation period during which demand/revenues are ramping up and may not be sufficient to reach viability of the mini grid projected in the pre-agreed business plan
 - Reimbursement during a pre agreed number of years starting not later than the date when the demand/revenues in the pre-agreed business plan are sufficient to achieve viability as reflected in the pre-agreed business plan
 - Triggering events to draw down expected to be set by reference to the verified level of the project company revenues below the agreed viability level in the audited business plan during initial (and riskier) stages of the operation of the sub-project
 - Triggering event for reimbursement expected to be set by reference to the verified level of the project company revenues above the pre-agreed viability threshold

Figure 9. Indicative structure for the contemplated reimbursable grants

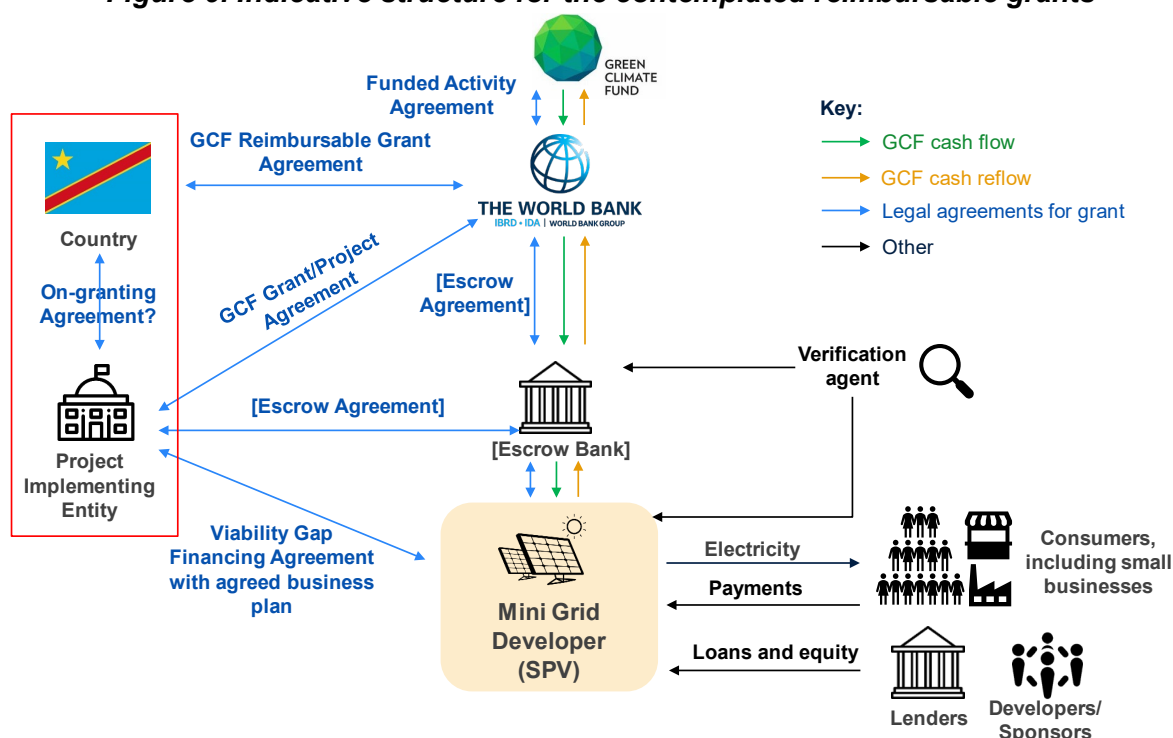
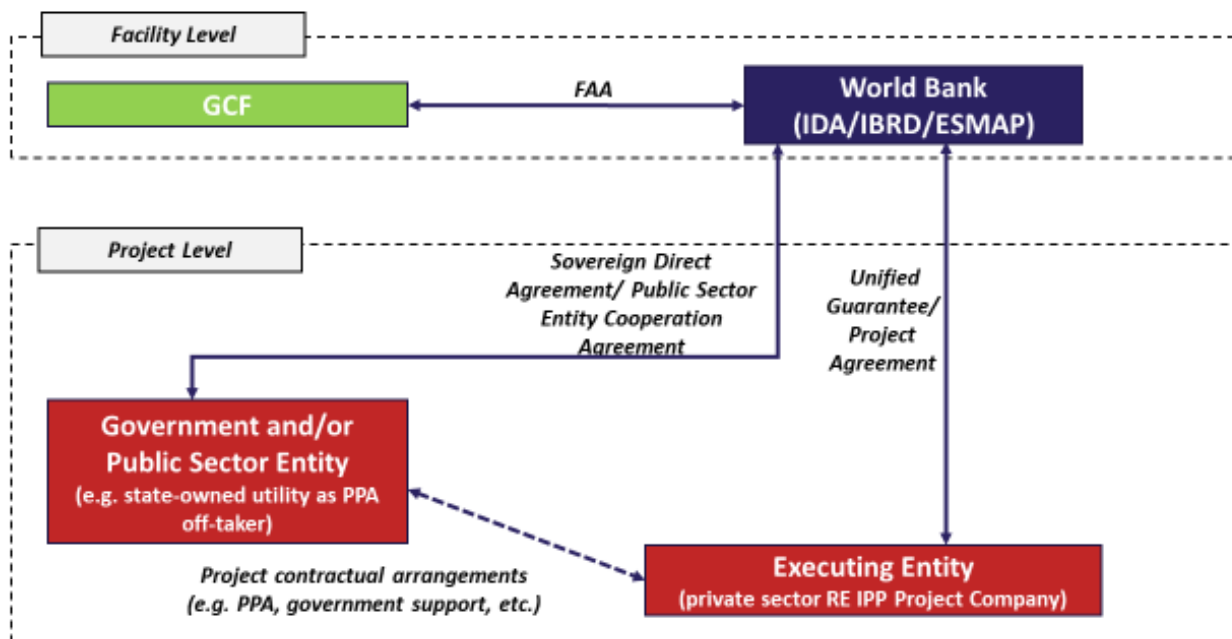
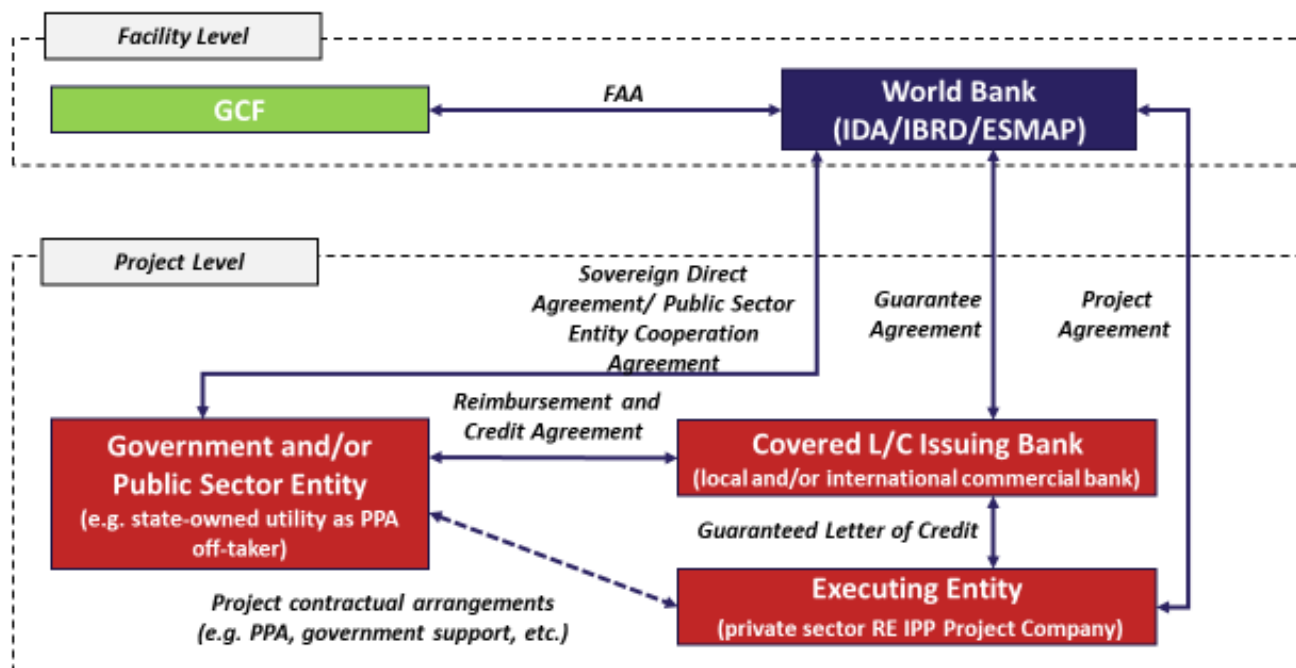


Figure 10: Indicative Implementation Arrangements for Component 3.2

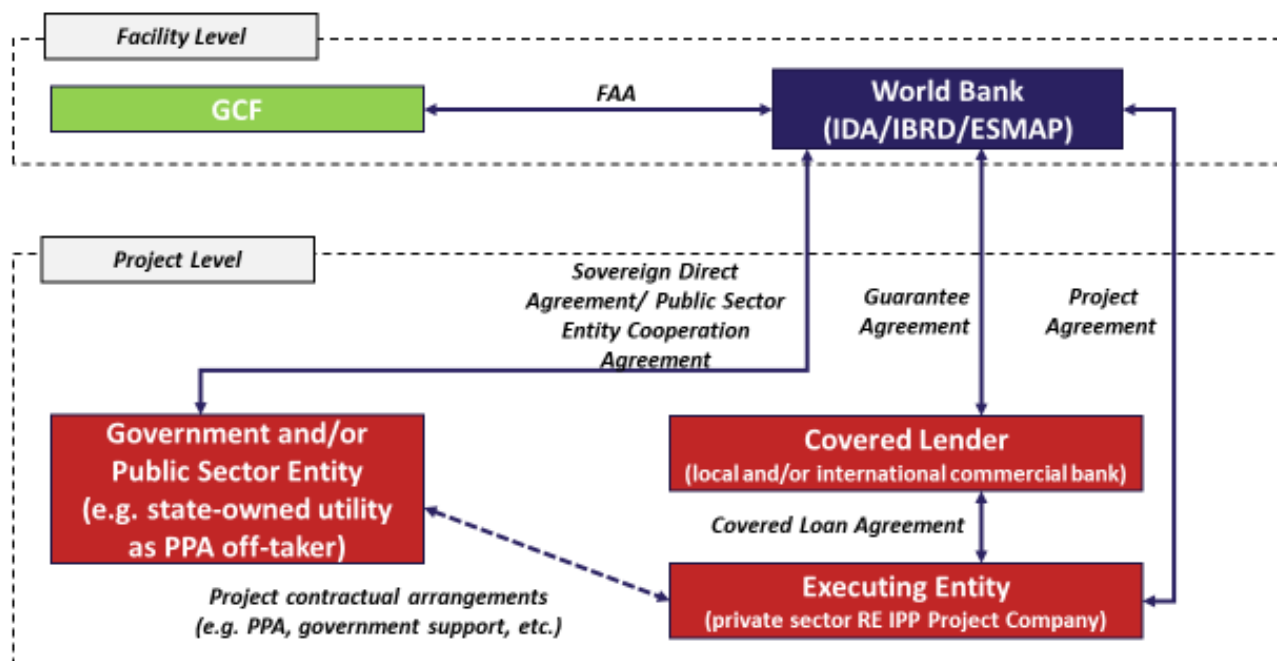
Direct payment guarantee for a private sector project (generic structure)



Payment guarantee (with a letter of credit) for a private sector project (generic structure)



Loan guarantee to a private sector project (generic structure)



Overview of the indicative mechanism contemplated for the guarantees:

- Eligible projects:** RE private sector projects competitively selected by the Host Country or a state-owned company mandated by the Host Country, with the technical assistance of the World Bank
- Executing Entity(ies):** private sector RE project company
- Beneficiaries:** (i) private sector RE project companies (in the case of the direct payment guarantee) or (ii) commercial international or local letter of credit issuing banks providing payment security to such private sector RE project companies (in the case of a letter-of-credit payment guarantee) or (iii) in the case of a loan guarantee, commercial lenders to the private sector RE project company.
- Covered Risk:** Two of the three structures (the letter-of-credit payment guarantee and direct payment guarantee) would support payments under a power purchase agreement (PPA) to an Executing Entity from a public-sector off-taker (i.e. liquidity risk), and the nature of the structure provided would depend on the project itself as it takes shape. The third structure, a loan guarantee, could provide appropriate support to commercial lenders (against public-sector risks related to long-term mobilization of private capital) to private investors (that is, Executing Entity RE project companies) for a RE project.
- Financial terms:** The conditions of guarantee would reflect World Bank's policy and procedural requirements but given the nature of guarantee projects, would be specifically negotiated with the participants in each project. In accordance with World Bank's policies and procedures, guarantees would only be provided to the extent necessary to mobilize private financing for the projects and/or to mitigate payment risks of the projects, taking into account country, project and market circumstances.

3. Projects Arrangements and Reporting Requirements

World Bank-financed projects follow a robust framework to design, prepare, implement, and supervise projects, as assessed by GCF prior to entering into its AMA with the WB. All the potential Facility-supported

projects will follow the WB project cycle, as well as the Bank's operational policies and procedures. For each Project included in the Programme, including those for which a GCF financing would be requested, the World Bank will carry out its due diligence and submit an appraisal-stage Project Appraisal Document (PAD) for quality control covering the IBRD/IDA financing and GCF financing, following standard WB procedures. Once all the relevant internal World Bank's requirements are met, the project moves to negotiations and World Bank Board approval stage, which cover the GCF financing and all WB or WB administered co-financing. The implementation of the Facility-supported projects (project implementation through completion and evaluation) follows the World Bank's applicable policies and procedures. For each project, a project management unit (PMU) will be established with a team expected to include at least a team lead, technical expert, procurement specialist, financial management specialist, environmental specialist and social/gender specialist, and a lawyer. Each project will have its own monitoring and evaluation (M&E) and reporting process, and the EE will be accountable for its implementation.

Figure 11: Key Personnel (Staff or Consultants) Typically Included in a Project Management Unit



B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

GCF funding is critical to address the current RE market failures identified and to ensure that countries continue their low carbon development pathway. Today, the deployment of solar and wind is constrained due to (i) lack of existing government strategy to adapt the current energy sector to the new disruptive technologies, (ii) weak grids that require upfront investments by public parties, (iii) government financing constraints, (iv) conflicting views on short-term vs medium-term economic growth needs, and (v) domestic production/import of fossil fuel. VRE deployment is also expensive when accounting the integration cost and when there is no organized competitive selection of IPPs, VRE deployment is not always competitive with coal generation. This may lead to a tough politico-economic decision when a government has to choose between promoting least-cost coal production versus slightly more expensive and more complex to integrate VRE as the case may be.

This situation is aggravated by the Covid-19 crisis which is impacting the RE deployment today. It is more difficult for the governments to spare public financing for critical public investments needed to unlock private investments as public financing is prioritized to tackle critical challenges hindering health and jobs. In addition, the utilities themselves are facing unprecedented challenges with payments issues and lower demand affecting their revenues, and therefore, private investors are perceiving them as riskier, increasing the premium on their cost of capital which in turn would increase the electricity price.

To face these challenges, the SRMI Facility is providing an integrated risk mitigation framework to (i) reduce the cost of privately-owned generation (lowering the risk premium embedded in the tariff) making

the electricity price more affordable, (ii) provide concessional funding leveraging climate finance to unlock critical public investments needed to enable private investments making them more affordable for the countries, and (iii) maximizing socio-economic benefits fostering jobs creation, skills development and women empowerment, leveraging the bidding processes.

Catalytic GCF concessional financing will be critical as (i) some countries among the poorest in the world are undergoing dramatic tensions on their budgets due to Covid-19; (ii) all will turn towards the cheapest source of energy with the lowest CAPEX under the current budgetary constraints excepts if it is not immediately impacting their budgets; and (iii) with the investment environment improving, concessionality is the last mile to raising private investors' interest. Under the current Covid-19 crisis and the risk to have coal and gas stranded assets supported by fossil fuel subsidies, climate financing and grant technical assistance to support developing countries in their energy transition are more critical than ever. Access to concessional public climate financing and DFI financing would enable solar and wind generation to be cheaper than the operating costs of most fossil fuel plants. In a context where the governments grapple with the critical task of dealing with the very real impacts of Covid-19 on the health and welfare of their population, countries may fall into a fast recovery strategy that would promote fossil fuel generation unless there is strong parallel support for renewable energy development. Therefore, countries need to be supported financially and technically to revise their current plans and implement the revised ones to avoid building new coal and gas plants. These are critical times to support new solar and wind projects in competing with the billions of USD spent on fossil fuel subsidies. In addition, by using concessional financing to reduce perceived risks by private sector investors, countries would be able to unlock at scale privately financed RE projects especially at a time where the utilities are weakened due to the financial situation of the countries. The GCF, thanks to its mandate and scale, will enable a scale-up of the SRMI approach that will enable real impact. In addition, supporting the scale-up of RE through a systematic, integrated and open approach such as SRMI will ensure the replicability in a timely manner of low-emission development pathways.

The use of climate concessional financing under the Initiative is also essential to support innovative technologies with a significant mitigation potential that are not yet commercially viable such as battery storage. GCF's risk appetite will be critical to provide innovative support to these countries. The industry supporting these paradigm- shifting disruptive technologies is at a nascent stage and will also suffer due to lack of market uptake unless there is continued demand from countries. Without additional investments in dispatch and system operation – including battery storage, VRE generation is likely to increase the risks of load shedding and systems' defaults. However, given the high cost of technologies and the risk arising from lack of experience in deploying these technologies at scale in the context of developing countries, private sector investors may not be willing to invest in these innovative technologies without development support and strong risk coverage.

Private investments in the off-grid RE space requires adapted risk mitigation instruments that can be developed with climate financing that is more flexible than DFI guarantees. A risk mitigation mechanism, as developed under Component 3.1, that covers critical risks such as demand off-take risk will be an important catalyst in helping attract more private sector investment and scale up mini grids and SHS. This demand risk is not covered by traditional DFI risk mitigation mechanisms like MIGA insurance or WB guarantees. GCF's support will be crucial to ensure that the right risk mitigation instrument is being provided.

With the exception of countries which receive full grants from the World Bank for instance due to their level of GDP per capita or post-conflict status (such as CAR in the current list of projects proposed to be supported by the SRMI Facility), the allocation between grants and loans is done between technical assistance/capacity building and infrastructure. IDA is providing grants to CAR

because it is an FCS country and is recovering from the recent major domestic political and security crisis in 2013. The ensuing insecurity led to massive forced displacement, the collapse of state institutions, and devastation of the country's economy. The country is in dire needs of IDA grant to recover its infrastructure and its economy. CAR remains among the poorest countries in the world and ranks second to last out of 188 countries on the United Nations Human Development Index. The other seven countries are split between three IDA and three IBRD. As per the SRMI methodology, the activities promoted under the technical assistance component are critical requirements for the development of a well-organized, sustainable and affordable generation mix as well as developing an enabling environment to private sector mobilization. These plans and studies are too expensive or complex in general for the countries to do without concessional support. Therefore, without targeted grant technical assistance, those studies are usually not being completed which leads to far from optimal decision with regards to new generation or transmission investments and they are usually not compatible with the current climate urgency. GCF's financing will be needed to ensure that the most fragile countries are able to reduce their debt weight and the cost of the interest.

Public investments are critical to enable the sustainable unlocking of private investments. However, governments are often stretched in their budget due to overwhelming social sector priorities. In this context, catalytic climate financing will help support more investments in RE. The approach promoted under SRMI targets private financing for the generation of renewables with the public investment being limited to critical public investment required for a higher penetration of renewables such as VRE integration investment, and solar parks infrastructures. However, governments need to continuously look at the budget allocation between different sectors, especially driven by pressing needs in the social sectors which is being exacerbated by the Covid-19 crisis. The possibility to access climate financing for those critical infrastructure investments, means that those countries would not need to reduce their IDA/IBRD envelopes for education or health sectors. The level of GCF participation in the financing of the currently proposed projects has been determined based on discussions with countries and their possibility to increase their financing for their access and RE generation agendas.

Another critical aspect of the SRMI Facility that will support an increased public investment allocation towards critical investments is the concessionality of GCF compared to IBRD and to a certain extent IDA as well. Under the SRMI Facility, GCF high concessional terms for vulnerable countries will be sought to be able to not increase the cost for the IDA countries while being attractive enough for the IBRD countries to make the investment towards low-carbon infrastructure fostering low-emission development pathways.

The GCF high concessional terms will enable the investments to be viable for both IDA and IBRD countries – and in turn the concessionality will fully benefit the electricity households of these countries. The GCF high concessional loan terms with a maturity period of 40 years and 10 years grace period with an all-in fee of 0.25 percent, i.e., 0.00 percent interest rate, 0.25 percent service fee (and waiver of 0.50 percent commitment fee) will result in an increase of 2 percent in project internal rate of return (IRR).

Given the abovementioned rationale, the GCF financing will be critical from the additionality perspective to unlock the sustainable programs presented by the seven countries included in this Funding Proposal from different perspectives, including (i) making the critical public investments needed more affordable, (ii) encouraging de-risking measures meeting the needs of the private sector through the availability of this concessional financing (e.g. mitigating the development risk for the private sector by financing solar/wind park infrastructures), (iii) adapting to the specific needs of some countries who need innovative risk mitigation approaches that are complex to provide by DFIs but that are within the mandate of the GCF and (iv) unlocking sustainable and bankable projects under SRMI at scale.

With the support of the GCF financing, the integrated risk mitigation framework promoted under the SRMI Facility is expected to play a key role on (i) a short-term basis by contributing to a green stimulus package for recovery and (ii) on a medium-term basis by ensuring a clean development path, mitigating significantly the GHG emissions trajectory.

B.6. Exit strategy and sustainability (max. 500 words, approximately 1 page)

Sustainability is at the heart of SRMI, mainstreamed in a cross-cutting integrated approach. Supporting countries to design and implement a sustainable low-emission development pathway, from a multi-dimensional perspective, fostering a transformational change with projects replicable at the level of the countries themselves:

- a. From a technical perspective, ensuring through comprehensive planning that sound RE targets are defined based on least cost planning, factoring in integration of VRE and grid resilience, and promoting a sector-coupling approach to the extent possible.
- b. From a financial perspective, reducing the reliance on public sector financing by supporting the design and implementation of bankable projects that could be financed by the private sector.
- c. From a strategy perspective, ensuring upfront that the RE program designed is aligned with the countries' priorities and strategic considerations (including in terms of risk allocation and roles and responsibilities of public parties).
- d. From an environmental perspective, financing countries' NDC in an economically viable manner while ensuring their implementation.
- e. From institutional and regulatory perspective, strengthening the countries' framework and developing the skills needed (including by financing the transaction advisors for the first round of projects under the RE program defined allowing 'a learning by doing' with hands-on experience), to give to the seven or more countries the ability to replicate the projects, and fully integrate the changes they are facing in our energy systems into their long-term strategy.
- f. From a socio-economic perspective, maximizing the positive impacts on the local industries, (positioning the local players on the solar/wind value chain in a competitive manner to the extent possible) and supporting local community development (through actions increasing their resilience and enhancing their livelihoods).
- g. From a gender perspective, ensuring that men and women benefit from the RE projects launched with a proactive policy to promote female employment throughout the RE project-cycle as otherwise men may benefit disproportionately from new employment opportunities across the RE value chain.
- h. From a financial perspective, SRMI's core investments in infrastructure for the grid and with regards to electrification are long-term investments that have long-term potential impact and will be critical to the economic development and transition to a new energy paradigm. The grid upgrades for VRE integration is expected to enable a shift on average from 7 percent penetration of VRE to over 35 percent by 2025 and electrification investments are core to the development of a population.

Therefore, the approach used by the SRMI Facility is supporting countries in developing a multi-dimensional sustainable RE program and operationalizing the first round of bankable projects. By laying strong foundations and implementing an adequate capacity building program, key public stakeholders

will be able to capitalize on the work done, the track-record built, and the skills developed for their next round of RE projects.

The SRMI Facility is expected to act as a catalyst to unlock these sustainable RE programs in the countries and pave the way to operationalize them through bankable projects privately financed. With a strong framework in place, a successful track-record and a sustainable ecosystem implemented, countries would be in a position to replicate these projects and to attract private investment in optimized conditions.

C. FINANCING INFORMATION							
C.1. Total financing							
(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)		Total amount			Currency		
		280			million USD (\$)		
GCF financial instrument		Amount	Tenor	Grace period	Pricing		
(i)	Senior loans	176	40 years	10 years	0.25%		
(ii)	Subordinated loans	n/a	n/a years	n/a years	n/a %		
(iii)	Equity	n/a			n/a %		
(iv)	Guarantees	20	25 years				
(v)	Reimbursable grants ⁶	30	15 years				
(vi)	Grants	54					
(vii)	Results-based payments	n/a					
(b) Co-financing information		Total amount			Currency		
		1,283.5 ⁷			million USD (\$)		
Name of institution		Financial instrument	Amount	Currency	Tenor & (grace) ⁸	Pricing	Seniority
World Bank (ESMAP)		Grant	1.5	million USD (\$)			
World Bank (IBRD)		Senior Loans	459.5	million USD (\$)	20 years	3.4%	
World Bank (IDA)		Senior Loans	750	million USD (\$)	30-40 years (5-10 years)	0.75% ⁹	
World Bank (IDA)		Grant	72.5	Million USD (\$)			
(c) Total financing (c) = (a)+(b)		Amount			Currency		
		1,563.5			million USD (\$)		
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)		CTF co-financing is considered for Mali to complement efficiently the GCF financing and benefit from the synergies between the CTF and the GCF. However, the CTF financing is not yet allocated for either projects. Therefore, it is not included in the co-financing section.					
C.2. Financing by component							

⁶ The reimbursable grant will be used as a risk mitigation instrument

⁷ The World Bank (IBRD/IDA/ESMAP) co-financing amounts are indicative and subject to approval by the World Bank Board and agreement with the EEs.

⁸ The pricing information for IBRD/IDA is indicative as the Bank Board periodically approves updated IBRD/IDA financing terms and they may change for a given country depending on when the project is submitted for Board approval. The prices presented in the table are for general IBRD and IDA projects.

⁹ The IDA pricing presented in the table refers to the standard service charge on IDA credits to SDR credits. As mentioned above, the pricing is indicative and additional interest charges and service charges may be imposed on borrowers depending on their categorization under the World Bank Directive on IBRD/IDA financing terms.

Component	Sub-Component	Activity	Indicative cost million USD (\$)	GCF financing		Co-financing		
				Amount million USD (\$)	Financial Instrument	Amount million USD (\$)	Financial Instrument	Name of Institutions
Component 1: Technical Assistance	Generation plans with VRE integration and Capacity Building (technical)	Generation plans with VRE integration and Capacity Building (technical)	5	5	Grants	0	Grants	Click here to enter text.
	Transaction Advisory and Capacity Building (procurement, legal)	Transaction Advisory and Capacity Building (procurement, legal)	8.5	7.5	Grants	1	Grants	World Bank (ESMAP)
	Solar and wind park preparatory studies	Solar and wind park preparatory studies	9.5	9	Grants	0.5	Grants	World Bank (ESMAP)
Component 2: Public Investments	Public investments in solar and wind park infrastructure	Public investments in solar and wind park infrastructure	126	28	Senior loans	98	Senior loans	World Bank (IDA/IBRD)
	Public investments for VRE integration grid upgrades and grid resilience	Public investments for VRE integration grid upgrades and grid resilience	479	128	Senior loans	351	Senior loans	World Bank (IDA/IBRD)
	Public investments for electrification	Public investments for electrification	771	20	Senior loans	751	Senior loans	World Bank (IDA/IBRD)
	[All sub-component 2 activities for FCS countries (CAR)]	[All sub-component 2 activities for FCS countries (CAR)]	90	30	Grants	60	Grants	World Bank (IDA)
Component 3: Risk Mitigation Instruments	Risk mitigation instrument for solar/battery mini-grids and off-grid private investors	Risk mitigation instrument for solar/battery mini-grids and off-grid private investors	30	30	Reimbursable grants	tbd	Guarantees	World Bank (IDA/IBRD)

	Risk mitigation instrument for grid-connected RE projects	Risk mitigation instrument for grid-connected RE projects	20	20	Guarantees	tbd	Guarantees	World Bank (IDA/IBRD)
Project Management	Project management	Project management	24.5	2.5	Grant	22	Grant/Loans	World Bank
Indicative total cost (USD)			1,563,500,000	280,000,000	1,283,500,000			

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?

Yes ☒ No ☐

C.3.2. Does GCF funding finance technology development/transfer?

Yes ☒ No ☐

Promoting a sustainable environment for private investments as part of the energy sector is critical to the SRMI approach. Capacity building is key from that perspective to ensure that local stakeholders are effectively empowered in relation to the solar/wind value chain to the extent possible. A multi-dimensional capacity building program is considered under SRMI for that purpose, covering from the design of the RE strategy to their implementation.

To increase the effectiveness of capacity building, the approach promotes a strategic analysis upfront to identify the targeted roles and responsibilities of the key stakeholders and assess the local industries' potential as to their positioning on the value chain. The objective being to define a concrete action plan to develop the targeted skills of the key stakeholders and improve their chances to benefit from their own RE projects. For example, by defining a cross-cutting country program to position the local players on the operation and maintenance of the solar/wind plants, allowing them to benefit from the transfer of knowledge needed as well as from the hands-on experience thanks to adequate arrangements reflected in the tender documents. This plan will benefit from the support of transaction advisors financed under SRMI to accompany the countries in this pathway throughout a 'learning by doing' approach for the pre-identified stakeholders.

In addition, as integration of VRE is a key aspect, SRMI is also promoting a transfer of knowledge in favor of key local stakeholders in relation to storage and more particularly with respect to battery storage. SRMI is leveraging the Energy Storage Partnership (ESP) launched by the World Bank in collaboration with partners to support this technology, not yet mature but presenting a significant potential to enable higher penetration of VRE in the grid. The SRMI Facility is directly contributing to the development of the battery storage technology by enabling the countries to implement it, building a track-record as well as crucial knowledge.

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's [Initial Investment Framework](#).

D.1. Impact potential (max. 500 words, approximately 1 page)

The seven proposed projects are expected to contribute to the shift to low-emission sustainable development pathways with the following:

- **Directly support the development of 2.5 GW of RE**, (i) 1.8 GW of solar PV, (ii) 400 MW of CSP, and (iii) 300 MW of onshore wind. Of this 2.1 GW are expected to be built during the lifetime of the projects and 0.4 GW expected to be built after the end of the projects. The direct support is enabled through Sub-Component 1.3 financing the solar and wind park analysis and Sub-Component 2.1 financing the public infrastructure of solar and wind park infrastructure and Sub-Component 1.2 for the hiring of transaction advisors to select IPPs. The SRMI Facility will also enable the integration of another 1.9 GW of RE. The integration of other projects is enabled thanks to the VRE integration analysis conducted under Sub-Component 1.1 and Sub-Component 2.2 under which core grid infrastructure are being financed.
- **Directly reduce GHG emissions by 97.93m metrics tons of CO₂**, more precisely 88.98m reduction for the 2.1 GW RE built under the projects over the 25 years of their operation, and 8.95m metric tons of CO₂ for 0.4 GW of RE projects that were prepared under the projects but not built during the lifetime of the projects¹⁰. The GHG emission reduction is based on RE projects being built.
- **Unlock USD 3.3 billion of private investments**, including USD 3 billion for the 2.1 GW built during the lifetime of the projects and USD 300 million for the 10.4 GW built after the end of the projects. The private investment is leveraged thanks to Component 1.2 transaction advisory and the general environment and bankable support provided to them through the general SRMI support of technical assistance to governments under Component 1, critical infrastructure under Component 2 and guarantees and risk mitigation instruments under Component 3.
- **Provide access to clean low-emission electricity to 4.2 million people thanks to the electrification investments**. Component 2.3 is critical in enabling public investments in clean energy mini-grids and connection to the grid that would have been made greener thanks to Sub-Components 2.1 and 2.2, and Sub-Component 3.1 that will enable unlocking private investments for green mini-grids.
- **Strengthen institutional and regulatory systems for planning and development.**
- **Reduce locking-in long-lived, climate-vulnerable infrastructure by increasing resilience of grid infrastructure.**

Table 3: SRMI Facility Proposed Projects Expected Outcomes

Country	VRE installed capacity (MW)	VRE projects prepared (MW)	VRE integration (enabled MW)	Battery Storage (MWh)	Private sector directly leveraged (mUSD)	Private sector enabled (mUSD)	Access (people)	Total GHG Emissions Reduced (Mtons/CO ₂ eq) during/after project
Botswana	300	0	0	50	820	0	600,000	52.3/0

¹⁰ GHG emissions calculations were made following the World Bank standards using the official combined margin grid emission factor per country was used.

CAR	35	0	0	35	10	0	250,000	0.4/0
DRC	100	0	0	140	200	0	1,500,000	0.06/0
Kenya	300	0	250	150	225	0	450,000	5.8/0
Mali	300	0	150	350	350	0	400,000	8.5/0
Namibia	400	0	0	200	940	0	1,000,000	8.4/0
Uzbekistan	600	400	1,500	50	450	300	0	13.5/9
Results	2035	400	1,900	975	2,995	300	4,200,000	89/9

The SRMI Facility aims to be scaled up to a total USD 400 million by adding countries, namely Indonesia and Vietnam, that would have projects following the agreed upon eligibility criteria as per the term sheet and the present funding proposal. The Phase 2 of the Facility – that is to be complementary to the current Phase – aims to – and such based on current project information - (i) avoid a total of 286.32 m metrics tons of CO₂, including 60.6 m metrics tons of CO₂ during project lifetime and 225.7 m metrics tons built after the end of the project, (ii) unlock a total of 6.1 GW, including in the short-term 2 GW and in the medium-term 4.1 GW of RE, and (iii) leverage a total of USD 10.2 bl, including USD 1.2 bl during the project lifetime and USD 9 bl built after the end of the project.

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

By tackling in a coordinated manner planning, procurement, development and curtailment risks to attract affordable private investments, SRMI aims at supporting countries in their shift to low-emission development pathways. Most countries have reached a point where most of their “easy” RE projects have been implemented. To move from a situation with a low VRE penetration to one where the majority of the electricity is produced from RE sources, governments will need to put in place sustainable generation plans, upgrade their grids and develop the right legal and procurement framework to leverage private investments at scale. This represents large investments from the public sector to enable such framework and will require strong governments buy-in and ownership. The SRMI Facility aims at accompanying governments in their path to low-emission development while ensuring it does not become an economic burden.

Building on a sound methodology¹¹, SRMI aims at supporting countries transform their strategy to low-emission development allowing them to harness their domestic resources at scale.

- The first step to a sound deployment is the setting of sustainable long-term targets for RE based on concrete analysis for demand, domestic resources and grid flexibility and stability. These targets will be the basis for a realistic but also ambitious plan to deploy RE in the given country. Through the SRMI Facility, the World Bank is supporting the seven countries to revise their generation plans and to identify the critical investments needed to integrate VRE into their grids while improving its reliability and resilience. These targets will then need to be integrated into their NDC revisions ensuring they are implementable.
- After the targets are set, the SRMI Facility is supporting the seven countries to develop a long-term strategy to implement those targets. The strategy looks at who and how to finance the deployment of RE and the associated infrastructure. It also identifies all the legal changes that need to be implemented to attract private investments and how the domestic lending sector can be leveraged to finance the debt part of the privately-owned power plants to reduce the risk of foreign exchange to

¹¹ “A Sure Path to Sustainable Solar : Solar Deployment Guidelines” was released during the 2019 United Nations Climate Summit in New York and are accessible online through the following link: <http://documents.worldbank.org/curated/en/244251575642432241/A-Sure-Path-to-Sustainable-Solar-Solar-Deployment-Guidelines>

the utility. In addition, socio-economic benefits can be integrated into the long-term strategy. It can support countries in leveraging the deployment of solar and wind to enable the population to gain opportunities for jobs and development.

- The first main pillar to the paradigm shift is capacity building. As part of SRMI, it is critical to increase capacity within the government on all fronts so that afterwards they will be able to conduct analysis without the support of consultants and to negotiate with the private sector on fair grounds.

After supporting the development of the country's long-term targets, its strategy to implement them and increase capacity, the SRMI Facility is financing the implementation its strategy. It is critical to countries to have a support which encompasses all from the target setting to its financing. Without such combined approach, the countries may not be able to finance the targets or have the right target in the first place.

As a consequence, countries will be able to operationalize their NDCs in a sustainable and timely manner in spite of the effect of the Covid-19 crisis, reducing significantly their GHG emissions trajectory by shifting to a lasting low-emission development pathway. Built on strong foundations through this comprehensive and integrated approach with a multi-dimensional sustainability perspective at its heart. This low-emission development pathway will be replicable, fast-tracking the road towards the Sustainable Development Goals for these countries.

The seven countries are representative of different types of markets/energy sector, with low to high electrification rates, low to high cost of generation, small to large grids etc. The success and lessons learnt from those seven countries will enable to replicate the SRMI approach across countries and enable to unlock sustainable RE markets for private investments. SRMI is currently being implemented/considered for another 30 countries in Sub-Saharan Africa and Asia which will benefit from the results and lessons learnt of this first phase.

Tailored support for each country will be provided under the SRMI approach. The status of RE deployment is very different from one country to another and will be further assessed at Project's level to leverage the existing related activities. The technical assistance deployed under the SRMI Facility will be tailored for each country needs and will, to the extent possible, build on the current RE activities deployed, in coordination with all stakeholders involved (including other DFIs as applicable) to enhance synergies. This will include for example adjusting existing least cost plans in place if needed or including the lessons learnt from existing RE projects in terms of risk allocation, government support, support to jobs' creation etc.

D.3. Sustainable development (max. 500 words, approximately 1 page)

Sustainability is at the heart of the multi-dimensional approach of SRMI. The SRMI approach aims at anchoring RE projects of the targeted countries in a sustainable ecosystem ensuring a lasting and replicable low-emission development pathway, fast-tracking in particular the SDG 13 Climate Action and the SDG 7 Affordable and Clean Energy.

Sound planning associated with adequate grid investments to enable a higher penetration of renewables while enhancing resilience will lead to the development of sustainable RE targets. It will avoid the lock-in of the countries in long-term high-emission development pathways. Harnessing domestic resources such as solar or wind for generation will also reduce the need for fuel imports and the potential economic impact of fuel cost fluctuations. It will increase the energy security of the countries, diversifying

their mix et improving their energy independence while reducing the price volatility (as electricity tariffs are set for a long-term period, around 25 years).

The operationalization of these RE targets through bankable projects with a fair risk allocation, building on a strengthened institutional and regulatory framework with adequate risk mitigation arrangements, aims to ensure the timely implementation of these projects. It aims to leverage private investments at scale, ensuring the availability of the commercial financing and the appetite of the private sector to take these projects forward. Hence, it will spare limited public sector finance for the countries which could be directed to other sectors in need such as healthcare, in particular in the context of COVID-19.

The strategic analysis upfront to define an action plan to maximize the socio-economic benefits of the RE program (embedded in the SRMI methodology), will be mainstreamed during the operationalization of the projects. It will enhance the involvement of the local players on the solar/wind value chain in a competitive manner to the extent possible, maximizing jobs creation, skills development and transfer of knowledge. It will also support local community development in the areas of implementation of the projects through actions increasing their resilience and enhancing their livelihoods. A specific attention will be brought to women empowerment as otherwise men may benefit disproportionately from new employment opportunities across the RE value chain.

Capacity building will be mainstreamed throughout the RE projects cycle, strengthening the countries' framework and developing the skills needed (including through 'a learning by doing' approach with the support of transaction advisors), to give to the seven countries the ability to replicate efficiently the projects. Building on the strong foundations of the program, key public stakeholders will be able to capitalize on the work done, the track-record built, and the skills developed for their next round of RE projects.

D.4. Needs of recipient (max. 500 words, approximately 1 page)

The seven countries have different economic development level and the Covid-19 crisis will impact their economic growth projections. GDP per capita is below USD 1000 per capita for CAR, DRC and Mali, between USD 1000 to 4,000 for Kenya and Uzbekistan and above USD 5000 for Botswana and Namibia. Their GDP growth is on average quite strong with the exception of Namibia that was hit with droughts in the last few years. The GDP growth forecast for 2020 will be reviewed with the current Covid-19 pandemic crisis. The countries also have quite different country investment ratings, with only Botswana considered as investment grade.

Table 4: SRMI Facility Proposed Countries Macroeconomic Overview

Country	Population (million)	GDP / capita 2018 (USD)	Inflation 2018 (%)	GDP growth 2017 (%)	GDP growth 2018 (%)	GDP growth forecast 2019 (%)	Country Investment Rating
Botswana	2.25	8,258	0.9	2.9	4.6	3.9	BBB+
CAR	4.7	475	1.6	4.5	3.8	3.0	n/a
DRC	84	562	29	3.7	5.8	4.4	CCC+
Kenya	51.4	1,710	4.7	4.9	6.3	5.6	B+
Mali	19	899	1.7	5.0	5.2	5.1	n/a
Namibia	2.5	5,931	4.3	-0.1	0.3	-1.4	BB+

Uzbekistan	32.9	1,532	17.5	4.5	5.4	5.6	BB-
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Source: World Bank, IMF, S&P

The seven countries have also quite different social development levels. The results of the Human Development Index (HDI) are also a good comparing instruments between the seven countries, with Botswana and Uzbekistan that have good HDI, Namibia and Kenya that have average HDI, and CAR, DRC and Mali with very low HDI.

Table 5: SRMI Facility Proposed Countries Key Social Development Index Overview

Country	Corruption Index (/180)	Human Development Index	Gini coefficient
Botswana	34	0.728	53.3
CAR	153	0.381	56.2
DRC	168	0.459	42.1
Kenya	137	0.579	40.8
Mali	130	0.427	33
Namibia	56	0.645	59.1
Uzbekistan	153	0.71	35.3

Sources: World Bank, Transparency International, United Nations

The seven proposed countries have clear future climate change impacts and a need to fundamentally adapt their energy systems by embracing disruptive technologies and transition to the grid of the future increasing resilience and sustainability while adopting low-development emission pathways. Further noting that Namibia and Botswana are not eligible for climate finance provided by the Climate Investment Funds (CIF).

SRMI aims at limiting the needs for public resources. Climate financing will be critical in providing sufficient financial incentives for those countries to change their business as usual way to tomorrow's energy system. This investment will be critical for countries to implement a clear strategy without impeding economic development or putting at risk their grid and therefore, their industrial development.

The countries considered will need strong support to strengthen their institutions and implementation capacity to transition to a disruptive technology grid. SRMI aims to provide upstream support to work with the countries on their technical and regulatory constraints and as a second step managing the implementation of their new strategy, such as with regards to managing battery storage into their grid, procuring IPPs and adapting their grid to electric mobility development.

D.5. Country ownership (max. 500 words, approximately 1 page)

While ensuring alignment with NDCs, relevant national plans, and/or enabling policy and institutional frameworks, Component 1 will be critical in providing inputs to the NDC/low-emissions development plans' objectives and regulatory frameworks and to enable competitive selection of IPPs with fair risk allocation. In each country, the World Bank has extensive engagement with relevant stakeholders, including national designated authorities. The proposed countries all have NDCs with the following commitments:

- Botswana:** The Government of Botswana has been spending a significant portion of its national resources to adapt to the impacts of climate variability over the years. The country intends to achieve

an overall emissions reduction of 15 percent by 2030, taking 2010 as the base year. The emission reduction target was estimated based on the baseline inventory for the three GHGs being CO₂, methane (CH₄) and nitrous oxide (N₂O). The reductions will be realized from the energy sources which is categorized as the stationary and mobile sources. The country is still working on developing initiatives for emission reductions in terms of long-term low carbon strategy.

- b. **CAR:** The annual emissions of the Central African Republic, estimated at 116,285.49 kt eq-CO₂ in 2010, or 26 tons eq-CO₂/person, represent less than 0.002 percent of global emissions. Despite its low rate of GHG emissions, the Central African Republic reaffirms its adherence to the principle of collective but differentiated responsibility and, in accordance with Decision 1CP/19, is taking the present ambitious measures to respond to this challenge without impeding its economic, social and environmental development. The Central African Republic aspires to reduce its emissions by 5% and 25 percent, respectively, in the 2030 and 2050 horizons in comparison to its reference BAU emissions and to increase its sequestration potential. With international support, it will emit around 33,076.1 kt eq-CO₂ less in 2050 than the annual reference emissions. The Central African Republic also aspires to reduce emissions of short-lived climate pollutants (SLCP), which science has shown have a significant short-term climate-warming potential and harmful effects on health, agriculture and ecosystems. Greenhouse gas emissions total 116,285.49 kt eqCO₂ with sectoral contributions of 89.46 percent for land use change and forestry; 5.26 percent for agriculture; 5.19 percent for energy (of which 4.91 percent is wood energy); 0.09 percent for waste and marginal amounts for industrial processes and use of solvents.
- c. **DRC:** As one of the most fragile states and poorest countries in the world, DRC faces a number of challenges which it seeks to address through its climate targets and focus on the SDGs. The contribution of the DRC to global GHG emissions is low (around 0.5 percent). Moreover, GHG intensity relative to GDP is also low. DRC has committed to reduce its GHG emissions by 17 percent by 2030, focusing specifically on three sectors – energy, agriculture and land use, land use change, and forestry (LULUCF) - which collectively produce nearly 98 percent of its GHG emissions.
- d. **Kenya:** Kenya's planning process on mitigation and adaptation hinges on the National Climate Change Action Plan (NCCAP) and the National Adaptation Plan (NAP). The two shall be reviewed every five years to inform the Kenya's Vision 2030 Medium Term Plan (MTP). The adaptation actions are further amplified in the NAP. Mitigation and adaptation actions are implemented across the various sectors at both the national and county government levels. The Ministry of Environment and Forestry coordinates the country's climate change affairs through the National Climate Change Secretariat (NCCS). The planning process follows the Climate Change Act (2016) that introduced several institutional reforms to enhance coordination of climate change adaptation and mitigation. These include the establishment of:
 - A high-level National Climate Change Council chaired by the President. The Council shall provide an overarching national climate change coordination mechanism and, among other roles, ensure the mainstreaming of climate change functions by the national and county governments, and approve and oversee the implementation of the NCCAP.
 - A Climate Change Directorate as the lead agency of the government on national climate change plans and actions to deliver operational coordination; and provide technical assistance on climate change actions and responses to County governments.

- Climate Fund to be a financing mechanism for priority climate change actions and interventions approved by the Council. In addressing climate change issues, public entities are required to undertake public awareness and consultations, and ensure a focus on gender, in line with the Constitution and the Climate Change Act (2016).
- e. **Mali:** The Ministry of Environment is responsible for adoption of NDC targets. There is no stakeholder involvement or multisector commission for the development of NDC targets. Currently, the country has no climate policies in place to mitigate climate change. However, Mali has requested technology transfer for means of implementation.
- f. **Namibia:** Namibia aims at a reduction of about 89 percent of its GHG emissions at the 2030-time horizon compared to the BAU scenario. The projected GHG emissions to be avoided in 2030 is of the order of 20000 Gg CO₂-eq. The Cabinet of Namibia is the Government entity responsible for approving policies. The Parliamentary Standing Committee on Economics, Natural Resources and Public Administration which usually advises Cabinet on relevant policy matters caters for the NDC. The Ministry of Environment and Tourism, which is responsible for all environmental issues in the country, is also the National Focal Point to the UNFCCC. It is the coordinating body for all climate change activities through its Climate Change Unit (CCU) of the Directorate of Environmental Affairs.
- g. **Uzbekistan:** In the long-term perspective up to 2030, Uzbekistan intends to strengthen measures and actions to struggle against climate change, in order to achieve the mitigation objectives: (i) to decrease specific emissions of greenhouse gases per unit of GDP by 10 percent by 2030 from level of 2010. Achievement of the INDC long-term objective envisages support from the international organizations and financial institutions, ensuring access to the advanced energy saving and environmentally sound technologies, resources for climate financing. There are currently some legally binding programs and acts aimed at ensuring by 2020 the implementation of measures for decrease in energy consumption, introduction of energy saving technologies in various sectors of the economy, social sphere and development of renewable energy: Decree of the President of the Republic of Uzbekistan, "On Measures for Further Development of Alternative Energy Sources"; Decree of the President of the Republic of Uzbekistan, "On Program of Measures for Ensuring Structural Reformation, Modernization and Diversification of Production for 2015-2019"; Resolution of the President of the Republic of Uzbekistan "On Program of Measures for Reduction in Energy Consumption, Introduction of Energy Saving Technologies in Economy Sectors and Social Sphere"; Program of Actions for Environmental Protection of the Republic of Uzbekistan; Resolution of the Cabinet of Ministers of the Republic of Uzbekistan, "On Integrated Program of Actions for Mitigation of the Aral Sea Disaster Impacts, Rehabilitation and Socio-Economic Development of Priaralie Region for 2015-2018"; Program for Further Development of Agricultural Production; Program for Further Irrigated Lands Improvement and Rational Use of Water Resources.

D.6. Efficiency and effectiveness (max. 500 words, approximately 1 page)

Since this is a Program level facility, economic and financial viability will vary from project to project. The Program efficiency and effectiveness is further presented in terms of cost-effectiveness, co-financing and leveraging, financial viability, and application of best practices.

▪ **Cost-effectiveness and efficiency**

The GCF will have a catalytical role to enable VRE at scale through the comprehensive and integrated risk mitigation framework promoted under SRMI by: (i) financing the critical public investments needed in an affordable manner and such by reducing the burden for the countries in particular in the context of the Covid-19 crisis where limited public financing is available; (ii) encouraging de-risking measures meeting the needs of the private sector through the availability of this concessional financing (e.g. solar/wind park infrastructure to mitigate the development risk); (iii) adapting to the specific needs of countries that need innovative risk mitigation approaches that are complex to provide by DFIs but are within the mandate of the GCF (e.g., to unlock large-scale mini-grids business models); and (iv) unlocking sustainable and bankable projects under SRMI at scale.

As described in B5, the concessionality of GCF financing will be critical to ensure that the countries are to implement those projects and avoid a path involving deployment of more coal and gas with limited RE. The cost of the public loans is a crucial parameter in the political decision to implement or not a more sustainable path. The GCF additionality is therefore paramount to a successful result. To change the course of a given political path, these countries are requesting financial support through their NDCs and as part of our ongoing dialogue. Therefore, the SRMI Facility is requesting the highest level of concessionality for the seven countries that are part of the Program.

The financial structure is a mix between grants for technical assistance, concessional loans for critical public infrastructure – with the exception of CAR which will get full grants due to its FCS situation – and risk mitigation instruments for private investors. Each financial instrument is critical in order to achieve the Program's objective to support countries in their low-emission pathways: (i) grants for key studies and capacity building will be critical to ensure that the right targets for RE are set and capacity is built within the government, (ii) concessional financing for public investments will support countries in increasing their penetration of VRE and improve grid reliability and resilience, and (iii) risk mitigation instruments are enabling the mobilization at scale of private investments. These are critical especially in countries where the counterfactual is cheap coal production such as in Botswana and Uzbekistan.

The Facility aims to unlock over 2.1 GW of RE in the short-term and 0.4 GW in the medium-term. The division between short-term and medium-term is mostly due to the planning for demand increase and preparation of parks. The expected GHG emission reduction is 89 million tons of CO₂eq which represents an average GCF dollar per GHG emission reduced of USD 3.15 and USD 17.57 if compared to the total public investments.

▪ **Co-financing, leveraging and leveraged long-term investments**

The USD 280 million GCF will be complemented by an expected USD 1.3 billion in World Bank financing which in turn is expected to leverage around USD 3 billion in private investments. The indicative leveraging between public investments (World Bank and GCF financing) and private investments is 1 to 3. The private investments are a result of the projects' public investments and therefore, are not considered as part of the co-financing and the Funded Activity.

▪ **Financial viability**

The GCF concessional financing will be critical to improve the economic and financial rate of returns of the projects. Keeping in mind that fuel taxes and subsidies are included in the economic analysis when data is available, the GCF financing increases the financial IRR on average by around 2 percent but

more importantly makes possible those projects that would not be accepted by the countries without the highly concessional climate financing support. GCF financing is a game changer for the governments as because it is concessional countries which would otherwise select polluting energy would be incentivized to join the program.

The financial analysis conducted for Botswana, Kenya, Mali, Namibia and Uzbekistan for Component 2.1 and 2.2 activities resulted in financial IRR of 13 percent, 18 percent, 27 percent, 12 percent, and 14 percent respectively. The level of concessionality proposed for each of these projects enables the financial IRRs to be financially viable for the countries.

▪ **Application of best practices**

To ensure that the best available technologies are considered, the government will hire transaction advisors that will develop the request for proposals to IPPs and ensure that the IPPs only select Tier 1 technologies. The best practice with regards to selection of IPPs will also be implemented to ensure financial sustainability of all the contracts and develop a track-record within the countries in selection of IPPs.

E. LOGICAL FRAMEWORK

This section refers to the programme's logical framework in accordance with the GCF's [Performance Measurement Frameworks](#) under the [Results Management Framework](#) to which the project/programme contributes as a whole, including in respect of any co-financing.

E.1. Paradigm shift objectives

- ☒ Shift to low-emission sustainable development pathways
- ☐ Increased climate resilient sustainable development

E.2. Core indicator targets

Provide specific numerical values for the GCF core indicators to be achieved by the project/programme. Methodologies for the calculations should be provided. This should be consistent with the information provided in section A.

E.2.1. Expected tonnes of carbon dioxide equivalent (t CO ₂ eq) to be reduced or avoided (mitigation and cross-cutting only)	Annual	3,559,000 t CO ₂ eq ¹²
	Lifetime	88,978,000 t CO ₂ eq
E.2.2. Estimated cost per t CO ₂ eq, defined as total investment cost / expected lifetime emission reductions (mitigation and cross-cutting only)	(a) Total project financing	USD 1,563,500,000
	(b) Requested GCF amount	USD 280,000,000
	(c) Expected lifetime emission reductions	88,978,000 t CO ₂ eq
	(d) Estimated cost per t CO₂eq (d = a / c)	17.57 USD / t CO ₂ eq
	(e) Estimated GCF cost per t CO₂eq removed (e = b / c)	3.15 USD / t CO ₂ eq
E.2.3. Expected volume of finance to be leveraged by the proposed project/programme as a result of the Fund's financing, disaggregated by public and private sources (mitigation and cross-cutting only)	(f) Total finance leveraged	4,278,500,000 USD
	(g) Public source co-financed	1,283,500,000 USD
	(h) Private source finance leveraged	2,995,000,000 USD
	(i) Total Leverage ratio (i = f / b)	15.3
	(j) Public source co-financing ratio (j = g / b)	4.6
	(k) Private source leverage ratio (k = h / b)	10.7
E.2.4. Expected total number of direct and indirect beneficiaries, (disaggregated by sex)	Direct	n/a
	Indirect	n/a
	For a multi-country proposal, indicate the aggregate amount here and provide the data per country in annex 17.	
E.2.5. Number of beneficiaries relative to total population (disaggregated by sex)	Direct	Click here to enter text. (Expressed as %) of country(ies)
	Indirect	Click here to enter text. (Expressed as %) of country(ies)
	For a multi-country proposal, leave blank and provide the data per country in annex 17.	

¹² The disaggregation of targets per country is provided in Annex 17.

E.3. Fund-level impacts

Select the appropriate impact(s) to be reported for the project/programme. Select key result areas and corresponding indicators from GCF RMF and PMFs as appropriate. Note that more than one indicator may be selected per expected impact result. The result areas indicated in this section should match those selected in section A.4 above. Add rows as needed.

Expected Results	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
	Tonnes of carbon dioxide equivalent (t CO ₂ eq) reduced as a result of Fund-funded projects/programmes	ex-ante and ex-post analyses, electricity generation data for individual power plants	0	7.12 ml t CO ₂ eq	28.47 ml t CO ₂	- Methodology: detailed in Annex 3 to the FP - Project lifetime: 25 years (which corresponds to the duration of the operation of the RE power plants by the private IPPs selected as expected to be reflected in the power purchase agreement to be signed) - Annual emission reductions: 3,559,000 t CO₂ eq - Lifetime emission reductions: 88,978,000 t CO₂ eq - Project implementation period: 12 years - Given the country investment project development is expected to take 1-3 years, the tender and IPP selection 1.5 years, and the construction is expected to 1.5 years, the average commercial operation date of the power plants is expected to start 5 years as from the
	Cost per tonne of CO ₂ -equivalent reduced	Third party report and ex-ante and ex-post analyses	0	17.57 USD	17.57 USD	
	Volume of public and private funds catalyzed by the GCF	Third party report	0	4,278,000, 000 USD	4,278,0 00,000 USD	
M1.0 Reduced emissions through increased low-emission energy access and power generation	M1.1 Tonnes of carbon dioxide equivalent (t CO ₂ eq) reduced or avoided - gender-sensitive energy access power generation	ex-ante and ex-post analyses, electricity generation data for individual power plants	0	7.12 ml t CO ₂ eq	28.47 ml t CO ₂	

						start of the project implementation period (should all the projects start at the same time). Lifetime costs of the programme: USD 1,563,000 (capacity building costs for USD 22.5 ml are included as they directly lead to GHG emission reductions) - Total financing is expected to be confirmed at project's stage (under the assumption that all the projects start at the beginning of the implementation period of the facility) and to be fully leveraged by the mid-term. At the stage of the Facility, it is expected as follows: ▪ Expected co-financing from the public source: USD 1,283,000,000 Expected private financing to be leveraged by the private sector IPPs selected during the tender process: USD 2,995,000,000 (expected to become available at the financial close of each project)
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E.4. Fund-level outcomes

Select the appropriate outcome(s) to be reported for the project/programme. Select key expected outcomes and corresponding indicators from GCF RMF and PMFs as appropriate. Note that more than one indicator may be selected per expected outcome. Add rows as needed.

Expected Outcomes	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
	<i>Number of technologies and innovative solutions transferred or licensed to support low-emission development as a result of GCF support</i>	ex-ante and ex-post analyses, project installation reports or any other source of information available	0 technologies	1 technology: BESS 487.5 MWh of BESS installed with dedicated capacity building	1 technology: BESS 975 MWh of BESS installed with dedicated capacity building	Technical studies confirming the needs of BESS and associated sizing Specific measures will be deployed under the Facility to support integration of variable renewable energy in the grid, among which the deployment of 1 technology the battery energy storage systems (BESS). 1 associated measure is expected: a dedicated capacity building covering VRE integration training in particular BESS
M5.0 Strengthened institutional and regulatory systems	<i>M5.1 Institutional and regulatory systems that improve incentives for low-emission planning and development and their effective implementation</i>	planning and strategic instruments developed in each country	0 systems	Adoption or adjustment (to the extent needed) of 6 integrated roadmaps to deploy RE projects through a competitive procurement scheme	6 RE roadmaps are operationalized through sustainable and bankable RE programs leading to a significant increase in the supply of RE by IPPS selected through the competitive procurement scheme put in place	The countries will have developed integrated roadmaps to harness their RE potential, updating their institutional and regulatory framework to the extent needed (in particular to allow for competitive RE tenders) and operationalizing these roadmaps into sustainable and bankable projects expected to be implemented by private IPPs selected through a competitive procurement scheme (except in some cases detailed in the FP such as in the case of CAR where a public procurement is contemplated given the FCV status of the country)

M6.0 Increased number of small, medium and large low-emission power suppliers	<i>M6.3 MWs of low-emission energy capacity installed, generated and/or rehabilitated as a result of GCF support</i>	ex-ante and ex-post analyses, project installation reports or any other source of information available	0	1,027.5 MW	2,035 MW	The studies under Component 1.3 are carried-out on time The implementation of the projects is completed as planned (no delay) The targets expressed in MW refer to the VRE capacity constructed (new grid-connected solar and wind power generation systems installed)
	<i>M6.2 Number of households and individuals (males and females) with improved access to low-emission energy sources</i>	ex-ante and ex-post analyses, household data, surveys, or any other source of information available	0 households 0 individuals, including 0 males 0 females	420,000 households 2,100,000 individuals, including 1,050,000 males 1,050,000 females	840,000 households 4,200,000 individuals, including 2,100,000 males 2,100,000 females	The pre-requisite studies are carried-out on time and the associated electrification plan is adopted on time The implementation of the projects is completed as planned (no delay) A total of 840,000 household representing 4,200,000 people out of which 2,100,000 women (precise numbers to be confirmed at projects' stage) is expected to benefit from access to electricity through different techniques of electrification (PPP mini-grids, grid extension etc.)

E.5. Project/programme performance indicators

The performance indicators for progress reporting during implementation should seek to measure pre-existing conditions, progress and results at the most relevant level for ease of GCF monitoring and AE reporting. Add rows as needed.

Expected Results	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
Sustainable RE targets available and presented in a government-owned least-cost generation plan	Evidence-based targets for low-emission pathways available in government-owned least-cost generation plan	Ex-ante and ex-post analyses, national plans and strategy documents, or any other source of information available	0	Sustainable RE targets in place in 3 countries under the Facility (3 plans with RE targets available)	Sustainable RE targets in place in 6 countries under the Facility (plans with RE targets available): Botswana, CAR, Kenya, Mali, Namibia and Uzbekistan	Least cost plans demonstrating the RE capacity targeted are in place and owned by the countries
Transparent and competitive selection of IPPs is in place	Successful competitive selection of IPPs	Ex-ante and ex-post analyses, procurement documents, or any other source of information available	Limited to no number of IPPs procured competitively	1 competitive process launched and successfully completed per country in 2 countries under the Facility (2 competitive biddings launched)	1 competitive process launched and successfully completed per country in 6 countries under the Facility (competitive biddings launched): Botswana, CAR, Kenya, Mali, Namibia and Uzbekistan including risk mitigation instruments proposed in Namibia and Botswana	The sub-components 1.2 and 1.3 are carried-out on time The design and bidding of the projects are completed as planned (no delay) Institutional and regulatory framework for competitive procurement of IPPs in place in the country Solar/wind projects procured competitively to select private IPPs in charge of the design, financing, construction, operation and maintenance of the plants with a bankable risk allocation Where needed the risk mitigation instruments requested by IPPs are provided, and they enable a reduction in the cost of PPA

Solar and wind parks ready for tendering	Solar and wind park critical studies are completed and the parks are ready for tendering	Ex-ante and ex-post analyses, construction completion reports, or any other source of information available	0	2 solar/wind parks are ready for tendering (2 sets of associated studies completed)	5 solar/wind parks are ready for tendering (5 sets of associated studies completed)	The sub-component 1.3 is carried-out on time RE projects tendered under a solar/wind park scheme (post completion of the critical solar and wind park studies)
Higher penetration of VRE	Enable an increase penetration of VRE in the grid	Ex-ante and ex-post analyses, energy statistics, country energy sector reports, studies or any other source of information available	% of VRE in the grid that can be integrated safely before implementation of the Facility projects (baseline values per country will be provided as part of the baseline studies)	Public investments in the grid enable a higher penetration of VRE to be safely integrated in the grid of 2 countries (2 countries with assessments showing a potential increase in VRE penetration from baseline)	Public investments in the grid enable a higher penetration of VRE to be safely integrated in the grid of all countries except DRC (countries with assessments showing a potential increase in VRE penetration from baseline)	No material changes to the least cost plans adopted RE projects deployed as planned Public investments for VRE integration grid upgrades implemented as planned Higher penetration of VRE is 15 percent and above of installed generation
Increased capacity by the Government and/or utility/dispatching entity with regards to VRE deployment and dispatch	Increased capacity of key staff designated by the EE	Ex-ante and ex-post analyses, surveys, or any other source of information available	Limited capacity in VRE deployment and dispatch by the Government and/or utility/dispatching entity	One training conducted in each country under the Facility	Two trainings conducted in each country under the Facility	Staff and experts have been designated by the EE for training and have adequate background and technical expertise to benefit from this capacity building and be able to apply the skills from the training in their operations Relevant staffs and experts designated by the EE have participated diligently in the training proposed
Private capital is leveraged for mini-grids in	Volume of private capital leveraged as a result of	Ex-ante and ex-post analyses, financial reports, or	0	0	USD 500 million	The studies under Component 1.3 are carried-out on time

DRC, and utility-scale RE in Botswana and Namibia	risk mitigation instruments	any other source of information available				The implementation of the projects is completed as planned (no delay) and the competitive selection of IPPs and mini-grid managers was successfully implemented
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E.6. Activities

All project activities should be listed here with a description and sub-activities. Significant deliverables should be reflected in the implementation timetable. Add rows as needed.

Activities	Description	Sub-activities	Deliverables ¹³
Component 1: Technical Assistance			
1.1.1 Generation plans with VRE integration analysis and capacity building (technical)	Least-cost generation and transmission plans are done jointly with VRE integration analysis and capacity building of technical planning teams and dispatch teams by each country EE	1.1.1.1 Preparation of a new (or adaptation of an existing) least-cost generation plan, which includes reiterative VRE integration analyses based on technical and commercial constraints, an evaluation of the dispatch capabilities and the limits of grid integration and determines a cost-optimized electricity mix that can meet demand at any time; 1.1.1.2 Preparation of a new (or adaptation of an existing) least-cost transmission/distribution plan, which includes capacity expansion models simulating generation and transmission capacity investment, given assumptions about future electricity demand, fuel prices, technology cost and performance, policy and regulations, and providing informed solar targets and indication of where best to locate future projects as well as a list of key investments needed to improve grid VRE integration capacity; and 1.1.1.3	A least-cost generation plan for 6 countries in the Facility (including grid investment needed to enable the targeted VRE) and at least two trainings per 6 countries, namely Botswana, DRC, Kenya, Mali, Namibia and Uzbekistan

¹³ Deliverables are indicative.

		Provision of institutional capacity building support for energy planning and dispatch (including using grid-connected battery storage).	
1.2.1 Transaction advisory and Capacity Building (procurement, legal)	Transaction advisory for IPP selection under competitive bidding, including legal and procurement capacity building by each country EE	1.2.1.1 Provision of technical assistance (including legal, procurement and transaction advisory services) and implementation of capacity building activities to support: (i) the assessment of the existing legal framework and the development of a legal framework for IPPs and competitive bidding conducive to private sector investment; (ii) the design of energy projects with fair risk allocation among stakeholders and contractual arrangements acceptable to potential investors; (iii) the competitive selection of investors; and (iv) the negotiations of the final contractual arrangement with the selected investors and closing.	Set of bidding documents setting the capacity of the solar and wind power plants to be installed by the private sector (Request for Proposals package of documents informed by the legal and structuring analysis done) available per country in the Facility, namely Botswana, CAR, DRC, Kenya, Mali, Namibia and Uzbekistan
1.3.1 Solar and wind park preparatory studies	Technical preparatory studies for solar and wind park infrastructure (i.e. feasibility studies, E&S instruments) by Botswana, CAR, Mali, Namibia, and/or Uzbekistan' EE	1.3.1.1 Preparation of preparatory and feasibility studies and environmental and social risk mitigation instruments, for the development of solar PV, solar PV with battery storage, CSP, onshore wind and offshore wind parks in support of investment de-risking and investor selection process. Preparatory and feasibility studies may include, as needed, geospatial analyses of the land around the substation, topography and geotechnical analyses, environmental and social analyses, site-specific	5 solar and wind parks ready for auction (technical studies completed) in the following countries: Botswana, CAR, Mali, Namibia, and/or Uzbekistan

		grid interconnection studies, solar irradiation/wind measurement analyses and other studies as needed.	
Component 2: Public Investments			
2.1.1 Public investments in solar and wind park infrastructure	Public investments in fencing, connection line, road, water access etc. for solar/wind parks by Botswana, CAR, Mali, Namibia, and/or Uzbekistan's EE	2.1.1.1 Carrying out infrastructure investments in support of the development of solar and wind parks, such as acquisition of land, construction of transmission lines and fencing; and	5 solar and wind park public investments completed in the following countries: Botswana, CAR, Mali, Namibia, and/or Uzbekistan
2.2.1 Public investments for VRE integration grid upgrades and grid resilience	Public investments for VRE integration grid upgrades (FACTS, transformer, doubling line etc.) and public investments for climate change grid resilience (vegetation control, doubling line, tsunami resilience etc.) by Botswana, CAR, Kenya, Mali, Namibia, and/or Uzbekistan's EE	2.2.1.1 Investment for grid modernization and upgrade to improve grid's flexibility to integrate larger amounts of VRE, limit curtailment or grid instability and adapt to climate change related shocks, including power system planning, grid management, strengthening of forecasting methods and grid code requirements. Grid reinforcements that support VRE integration may include (i) addition or replacement of lines and transformers for grid extension and capacity enhancements (both for answering growing demand and for integrating VRE power), (ii) equipment for smoothing the voltage and frequency issues, such as capacitor banks, battery storage and other reactive power compensators, together with flexible alternating current transmission systems, (iii) equipment for faster and more efficient grid operation, such as monitoring systems, demand and production forecasting systems, and automats for controlling generation units and grid	Reinforced grids enabling further safe VRE integration and grid more resilient to climate change in the following countries: Botswana, CAR, Kenya, Mali, Namibia, and/or Uzbekistan

		operations through automatic generation control; and (iv) other innovative technologies whose commercial viability may not have been established yet such as load shifting battery storage.	
2.3.1 Public investments for electrification	Public investments for electrification (PPP mini-grids, grid extension etc.) by Botswana, CAR, DRC, Kenya, Mali and Namibia's EE	2.3.1.1 Investments in solar home systems (SHS), mini-grids and grid connection, as best appropriate to the country context, with a particular focus on women.	Supply of electricity through grid expansion, mini-grids and/or SHS to 840,000 households across 5 countries: Botswana, CAR, DRC, Kenya, Mali and Namibia

Component 3: Risk Mitigation instruments

3.1.1 Risk mitigation instruments for solar/battery mini-grids and off-grid private investors	Risk mitigation instruments for RE mini-grids and off-grid private investors by DRC's EE	3.1.1.1 Technical assessment (informed to the extent possible by a market sounding) to ensure that the coverage proposed meets the needs of the private sector investors, allowing long-term mobilization of private capital matching the needs of the projects	Identification of the terms and conditions of the reimbursable grants available for inclusion in the sets of bidding documents for RE mini-grid selections in DRC
3.2.1 Risk mitigation instruments for grid connected RE projects	Risk mitigation instruments for RE grid connected projects by Namibia and Botswana's EE	3.2.1.1 Technical assessment (informed to the extent possible by a market sounding) to ensure that the coverage proposed will meet the needs of the private sector investors, allowing long-term mobilization of private capital matching the needs of the projects.	Identification of the terms and conditions of the guarantees available for inclusion in the sets of bidding documents for RE competitive bidding in Namibia and Botswana

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

Besides the arrangements laid out in the AMA between the World Bank and the GCF, the implementation of the SRMI Facility involves (i) monitoring of program performance indicators (as per Section E5); (ii) periodic progress reports; and (iii) midterm review of implementation progress. It is noted that the end targets of the Programme are best estimates as it is not known upfront exactly the results of each project that will be financed, and the targets may be revised during AE's due diligence, project preparation and the mid-term review as necessary. The AE will report to GCF the implementation status at both Facility and country level by aggregating specific country-level information and data.

▪ **Monitoring**

Specifically, the implementation of each project under the SRMI Facility will be managed and monitored at project and Facility level by World Bank's in-house staff with inputs from, as applicable, Project Implementation Units (which may include staff and consultants). The World Bank has dedicated staff in its energy team, operation departments, Environment and Social departments and country offices that will conduct due diligence, monitor compliance and performance risks, implementation of the Environmental and Social Commitment Plans and the stakeholder engagement plans.

The World Bank will follow its standard monitoring and reporting policies and procedures, which will include semi-annual reports on the implementation status and performance of each project. By the end of each project supported by the SRMI Facility, the Bank will conduct jointly with each respective Executing Entity an implementation completion assessment and prepare an implementation completion report (ICR) which will review the performance of the project, assess effectiveness and efficiency of project implementation, the achievement of the project development objective, and provide relevant lessons learned. Financial reporting is further discussed in Section G3.

▪ **Reporting**

Reporting of EEs to World Bank. As specified in the subsidiary agreements between the EEs and the World Bank, the EEs are obliged to report on the implementation of the Project and the use of proceeds of the Facility in accordance with the terms and conditions of the subsidiary agreements, including on the environmental and social performance of the project and the achievement of the indicators agreed between the AE and the EE to the World Bank on an annual basis. For each project supported by the SRMI Facility, the respective EE will be responsible for overall M&E of the project, including collection of project performance information and reporting on project impacts and results.

Reporting of World Bank to GCF. The World Bank will provide to the GCF (i) annual activity performance reports (APRs) on the status of the Facility throughout the relevant reporting period, (ii) mid-term evaluation reports at the midpoint of the implementation period of the Facility and (iii) final evaluation reports at the end of the implementation period of the Facility as per the FAA provisions. The APRs will provide information specific to projects supported under the Facility, as relevant.

▪ **Evaluation**

The evaluation methods for mid-term and final evaluations will include:

- a. Key informant interviews with relevant beneficiary staff and clients, as well as other key stakeholders, based on stakeholder mapping; and
- b. Desktop review of relevant Facility documentation, including Project Implementation Unit reports, based on an agreed list of evidence to be provided by the client.

Please refer to Annex 11 for further details.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

The main risks identified to the successful implementation of the Facility are linked to Governance (political and governance, macroeconomic and sector strategies and policies), Technical and Operational (technical design of projects, institutional capacity for implementation and sustainability, fiduciary and stakeholders), uncertain depth and duration of Covid-19 pandemic (with direct impact on macro-economic environment and possibly project preparation and implementation context in various countries) as well as Security in specific project countries.

Selected Risk Factor 1

Category	Probability	Impact
Governance	Medium	Medium

Description

Political and Governance. As DRC, CAR and Mali are facing complex political situations, it may impact their political commitment to the projects and the governance aspects of the implementation of the components:

- DRC is still recovering from a series of conflicts that broke out in the 1990s, creating a protracted economic and social slump. Joseph Kabila served as head of state between 2001 and 2019. Presidential and parliamentary elections, initially slated for November 2016, were postponed until December 30, 2018 to facilitate an overhaul of the electoral register. Felix Antoine Tshisekedi—the son of Etienne Tshisekedi, the country's longstanding opposition leader—secured 38.5 percent of the vote and was declared the winner by the national independent electoral commission. This victory was contested, including by the National Episcopal Conference of Congo (CENCO), which had deployed 40,000 election observers.
- CAR's elections in 2016 brought an end to three years of political transition and turmoil. Since 2016, CAR has been governed by President Faustin-Archange Touadéra and Prime Minister Simplicie Sarandji. The next presidential elections are scheduled to be held between December 2020 and March 2021.
- Mali has been experiencing instability and conflict since the military coup of 2012 and the occupation of the northern regions by armed groups. The operations of the United Nations Multidimensional Integrated Stabilization Mission in Mali (MINUSMA) date back to July 2014.

Macroeconomic. This risk category includes external and domestic macroeconomic risks that may derail the achievement of operations development results during either preparation or implementation. Overall, the seven countries are quite stable from a macroeconomic perspective. GDP growth in 2017 and 2018 has been negative in Namibia, between 2 and 4 percent in Kenya and Botswana and above 5 percent in DRC, Mali and Uzbekistan. Their inflation levels are aligned with their GDP growth levels with rates in 2018 between 0 to 5 percent. However, it is important to note that it is currently unknown how the covid-19 crisis will impact their economies. The emerging view in the case of Kenya is that the Covid-19 crisis is having a large negative impact on the economy, there has been a significant widening of the fiscal and external financing gaps and consequently, GDP growth has been downgraded.

Sector Strategies and Policies. It is critical to have in place the right regulatory framework: (i) Kenya are working on new legislation to implement competitive selection of IPPs, (ii) DRC does not have the full policy in

place to enable privately-owned mini-grid operations, and (iii) Botswana, Namibia and Mali are missing clear generation plans with implementation strategies linked to it.

Mitigation Measure(s)

The World Bank team is working closely with the different Governments to ensure the sustainability and relevance of the projects to the given country. In general, good governance of the projects will be ensured by the adoption of transparency, accountability and participation principles and attention to fraud, corruption and other unethical practices resulting from government failures. These will be enabled thanks to the World Bank implementation and M&E systems.

The projects are expected to strengthen the country's macro-fiscal situations as through the deployment of least-cost renewable energy it will reduce the cost of imported fossil fuel while reducing energy security risks and improving national capacity.

In addition, the World Bank is working with the seven countries to ensure that a comprehensive regulatory framework is in place when the new investments come in play aiming to ensure the sustainability of the new generation projects.

Selected Risk Factor 2

Category	Probability	Impact
Technical and operational	Medium	Medium

Description

Technical Design of projects. As solar and wind generation are well known technologies and there are clear investments related to VRE integration, the technical design risk is minimal. In addition, with a clear and fair risk allocation between the private and the public stakeholders, RE projects are least-cost compared to other generation projects.

Institutional Capacity for Implementation and Sustainability. The institutional capacity of the implementing agencies, implementation arrangements, and M&E arrangements is quite different from one country to the next. Indeed, Namibia, Botswana, Kenya and Uzbekistan have strong implementing agencies whereas CAR, DRC and Mali have weaker institutional capacity.

Risk of RE Projects Not Materializing. Most RE power plants under the Facility will be prepared thanks to public investments. However, the financing for the power plants themselves will be financed by private investors. The private investment is not covered under the Facility.

Mitigation Measure(s)

To ensure that this risk is mitigated, the World Bank is supporting the proposed seven countries in their selection of the private investors. For the countries with limited capacity, larger capacity building components will be provided with additional external support for project implementation. A transparent and well-organized selection of private investors combined with the reduction in risks from curtailment, land and E&S perspective

thanks to the park scheme will ensure that private investment is indeed leveraged as a result to the Facility's projects.

Each country has a different fiduciary risk and it is being mitigated by previous experience with each client, and World Bank procurement and financial management requirements.

Selected Risk Factor 3

Category	Probability	Impact
Other	Medium	Medium

Description

Security. Security risks are key in countries such as Mali, CAR and DRC:

- Mali's security, which is critical for economic recovery and poverty reduction, remains fragile in the face of continued attacks by armed groups on UN peacekeepers, the Malian army, and civilians, mainly in the north and central regions. In July 2017, Germany, the African Development Bank, the World Bank, France, the European Union, and the United Nations Development Programme launched the Sahel Alliance in order to provide a coordinated and tailored response to the challenges faced by the G5 Sahel member countries (Burkina Faso, Chad, Mali, Mauritania, and Niger). Since that time, Denmark, Italy, Luxembourg, the Netherlands, Spain, and the United Kingdom have joined the Alliance.
- On February 6, 2019, the Government of CAR signed an African Union-mediated peace agreement with 14 armed groups. This agreement has been endorsed by the international community. The African Union and the Economic Community of Central African States (ECCAS) serve as guarantors of the agreement, while the UN peacekeeping force (MINUSCA) plays a critical support role in the background.
- DRC is fraught with political instability, armed clashes, and human rights violations. Conflict erupted in 2016 in the Kasai region, which includes five provinces in the center of the country. It is another instance of fighting between the military and splintered ethnic militias. Nationally, 2.1 million people were newly displaced in 2017 and 2018, making the DRC the African country with the highest number of internally displaced people — 4.5 million.

Mitigation Measure(s)

The security risk will be monitored, and the projects will rapidly adapt to any change in security levels in the three countries.

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

The World Bank has a new Environmental & Social Framework (ESF) that will be used by all projects.¹⁴ The ESF is divided in ten E&S standards (ESS) that are each assessed based on their relevance depending on the project's concept. The main aspects considered at this stage from an E&S perspective are summarized below. However, each project will have its own assessment completed during project preparation. As a few of the projects aim to finance 500kV lines to unlock GWs of VRE, these projects as part of the Facility will be Category A following the GCF classification and may be classified as high or substantial risk as per the World Bank ESF classification.

▪ **ESS1: Assessment and Management of E&S Risks and Impacts**

The projects are expected to bring substantial direct environmental benefits by increasing the deployment of renewable energy. Transitioning to a higher percentage of renewable energy in the electricity mix has benefits for countries with regards to GHG emissions, air and water pollution, and use of water resources. The projects are likely to result in significant employment generation, including construction jobs, followed by a smaller number of long-term jobs associated with operation and maintenance (O&M) activities. There are substantial opportunities to promote female employment (direct and indirect) through training and incentives.

If not properly managed, the downstream development of the infrastructure for the solar/wind parks, as well as the construction and operation of the RE power plants may result in potential land conversion (and potentially degradation), habitat loss, excessive water use, discharge of hazardous wastes in manufacturing of PV solar cells and wind turbines and disposal of solar cells, and risks to worker and community health and safety. However, most of these potential impacts are low to moderate in intensity/significance and are reversible and localized in nature, and therefore can be easily prevented, e.g. by aligning site selection with ESF requirements, and mitigated, e.g. by sound operational practice during operation.

Under ESS1, an E&S Commitment Plan (ESCP) will be required, and it will specify all other E&S plans and instruments that would be required for each project. The ESCP is a legally binding document that is referred to in the legal agreements between the Bank and the EEs/implementing agencies.

▪ **ESS2: Labor and Working Conditions**

Project workers are expected to be mobilized under all the projects. The EEs/implementing agencies will have to comply with the ESS2 requirements as presented in Labor Management Procedures (LMP) to be prepared for each project.

▪ **ESS3: Resource Efficiency and Pollution Prevention and Management**

The downstream risks and impacts related to potential water scarcity (also considering seasonal water availability and any cumulative effects by multiple demands), the release of pollutants, waste generation, the management of disposal materials and hazardous wastes, impacts on communities, and resource use

¹⁴ Depending on the final design of the projects to be financed out of Component 3, the World Bank Group Performance Standards may apply to some of the financed activities under these projects, in accordance with the World Bank's relevant policies and procedures.

efficiency will be assessed for each project. However, most of these potential impacts are low to moderate in intensity/significance and are reversible and localized in nature, and therefore can be easily mitigated.

During the construction phase of the solar and wind parks, there may be some potential negative environmental and social impacts including, air quality deterioration due to dust and exhaust emissions during construction activities. These impacts require appropriate assessments and management measures to contain them.

▪ **ESS4: Community Health and Safety**

ESS4 requirements will be taken into account in the implementation of project's activities. While the construction and operation of solar/wind power plants has risks accompanying both normal operations and potential accidents, there have to date not been any accidents of the severity that have occurred in other forms of energy production (e.g. in hydropower). There is a possibility of noise emissions from the running of construction machinery and related transportation activities, community nuisance etc. These impacts require appropriate assessment and management measures to contain them.

▪ **ESS5: Land Acquisition, Restrictions on Land Use and Involuntary Resettlement**

Land needs for solar and wind parks are substantial. Any potential land acquisition or land use restriction will be assessed and managed in a manner consistent with ESS5 requirements, in order to avoid, minimize, reduce/mitigate and compensate at replacement cost land acquisition based on due diligence and plans prepared in accordance with ESS5.

▪ **ESS6: Biodiversity Conservation and Sustainable Management of Living Natural Resources**

The proposed solar/wind projects, if not properly designed and sited, may result in loss, degradation or fragmentation of habitats and other adverse impacts on biodiversity and living natural resources. Care will be taken to ensure that the impacts of such projects on natural habitats, terrestrial and aquatic ecosystems and species are well assessed and managed.

▪ **ESS7: Indigenous Peoples/Sub-Saharan African Historically Underserved Traditional Local Communities**

The presence of indigenous people in the project areas or their collective attachment to such areas will be screened and confirmed by the World Bank in consultation with the national authorities. Once it is confirmed, consistency with ESS7 objectives and requirements will be integrated into the design of the project and E&S documents, including the development of an indigenous peoples' development plan or other types of plans depending on the country and project context, and Free, Prior and Informed Consent in the circumstances specified in ESS7.

▪ **ESS8: Cultural Heritage**

Each project will screen the project area and assess potential impacts on cultural heritage, both tangible and intangible. The site selection for each project will be mindful of cultural heritage and the application of ESS8 will be further assessed under each project.

▪ **ESS9: Financial Intermediaries**

The projects are not expected to be financial intermediary projects.

▪ **ESS10: Stakeholder Engagement and Information Disclosure**

Project stakeholders can include communities or individuals affected by the project and their formal and informal representatives, national or local government authorities, politicians, religious or community organizations and civil society groups with special interest, academic communities, and businesses. The identification of project-affected parties (individuals or groups) will also include those who, because of their circumstances, may be disadvantaged or vulnerable; i.e. those who may be more likely to be adversely affected by the project impacts and/or more limited than others in their ability to take advantage of the project benefits.

A Stakeholder Engagement Plan (SEP) will be developed for each project, to ensure transparency and meaningful consultation with the project-affected and interested parties. Stakeholder engagement and consultations will be conducted throughout the project cycle. This will include discussions of project design and impacts as well as multi-stakeholder discussions on these issues during the preparation phase. The SEP, along with other E&S instruments, will be subject to public consultation and disclosure per requirements of ESS10 and will be treated as a living document to be regularly updated as needed during project implementation.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

The Facility aims to enhance female economic participation in the energy sector and create income-generating opportunities for women in project countries. Proposed interventions under the Gender Action Plan are both aligned with the World Bank Group Gender Strategy (2016-2021): (i) advancing equality between men and women through climate change mitigation and adaptation actions, and (ii) minimizing gender-related risks in climate change actions. Employment opportunities created by large-scale solar and wind infrastructure projects often benefit men more than women, resulting in a significant employment gap in the RE industry. Female access to electricity can at times also be limited in target countries, and specific attention to women's connectivity will be key in ensuring men and women equally gain from energy services. Under the SRMI approach, issues related to female employment and access to electricity will be effectively embedded across the RE project-cycle, from the procurement of the RE project to its implementation, going beyond whenever possible to also enhance livelihood in affected areas.

These issues will be tackled in a sustainable manner, raising awareness, improving gender policies, building capacities and generating knowledge on the program's differentiated impact on men and women. Preliminary gender assessments for each of the seven countries under the Facility highlighted the following gender gaps and their potential mitigation actions:

- a. **Female employment and skill gaps in the energy sector:** the main barriers to women's participation in the energy sector include social norms that confine women to unpaid household and care work, occupational gender segregation, limited capacities and skills, as well as discriminatory work environments. Worldwide, the RE industry employs only about 32 percent of women. Therefore, without a pro-active focus on institutional reform to promote female employment gender-policy, men may benefit disproportionately from new employment opportunities across the RE value chain,

particularly in construction, operation and maintenance. This gap will be addressed through skills building and, outreach efforts, as well as the promotion of equal opportunities practices at the corporate level (including through programs to afford women the ability to gain hands-on professional experience).

- b. **Female employment in the areas of implementation of RE projects:** RE projects are frequently located in remote areas with good solar/wind resources and large plots of lands, but where living standards are below the country average. Women, if employed, are more exposed to lower-paid, lower-quality employment in the informal sector. Under the SRMI approach, the socio-economic needs of the local communities will be assessed, and actions empowering women prioritized. Those may include investments in education and skill-building, for instance in traditional home-based activities such as handicrafts, or in agricultural activities. Special attention will be paid to the promotion of local women-run businesses along the project development cycle.
- c. **Gender-based violence:** women routinely experience sexual harassment and intimate partner violence in target countries, and the risk of gender-based violence may increase during the realization of large-scale RE projects. This risk will be mitigated through implementing mitigation and response approaches and the adoption of internal prevention and response mechanisms in compliance with the World Bank Group GBV practices. These interventions are expected to efficiently complement actions empowering local women and men and aim to create a safe environment at the project site and in communities.
- d. **Women access to services as energy consumers:** among project countries with lower electrification rates and under-developed grids, women suffer from time-poverty and increased health-issues, as they bear the responsibility of food preparation, as well as fuel and water collection. Limited access to electricity among poor households, including female-headed households (FHH) can also be explained through affordability constraints regarding electricity connection or use of energy services. Improving access to electricity services (off-grid and on-grid) may reduce the time and labor burden of women, generate health and education benefits, increase female labor participation and incomes. For these reasons, the EEs will be required to (i) pay attention to women and men's differentiated needs in the design of electrification programs, (ii) conduct inclusive stakeholders' consultations to ensure women's voice is heard and included in decision-making processes, (iii) undertake outreach and capacity-building initiatives to increase women's knowledge about new forms of energy, and (iv) prioritize FHH in the design of actions to assist poorer households with connection charges.

The Facility will follow a systematic approach and provide cross-support to projects interventions with a focus on data collection, policy engagement, project level actions and capacity-building activities. The WBG ESMAP Gender and Energy Program will be available to provide assistance to the SRMI facility, as part of its mandate to strengthen women's role in the energy sector as consumers, employees and entrepreneurs. The Facility will encourage the recruitment of a balanced female-to-male employment ratio within the PMU team.

Obtaining stakeholders buy-in will be key in ensuring that interventions to promote women's employment and enhance their livelihood are effectively implemented with measurable outcomes. Countries adherence to international agreements and national normative frameworks on gender equality,

will be a good starting point to strengthen government's support. At the project and program level, the following approach will be followed:

a. Data collection and initial assessments:

- Build upon existing country gender diagnostics and action plans;
- Document the baseline situation and identify gaps through collection of sex-disaggregated data;
- Conduct stakeholders' consultations involving female participants, to understand men and women differentiated needs; and
- Include an analysis of the differentiated impact of the SRMI project on men and women in initial assessments, through the involvement of gender and energy experts (Environmental and Social Impact Assessment, Resettlement Action Plans etc.).

b. Identifying and implementing relevant actions:

- Identify and implement relevant interventions to narrow gaps in terms of employment and energy access, based on the initial diagnosis and inclusive stakeholders' consultations; and
- These may include (i) actions to promote female employment in the renewables sector (including capacity building and programs for women to gain hands-on experience in the energy sector such as support for school-to-work transition), (ii) financing of local communities projects empowering women in the area of implementation of the RE projects (such as activities generating revenues), (iii) implementation of prevention measures to mitigate GBV in the area of implementation of the RE projects and (iv) facilitating of the energy access for FHH).

c. M&E and Knowledge management:

- Monitor results throughout the project cycle to assess impact on men and women and allow for real-time adjustments; and
- Expand the knowledge base by documenting lessons learnt from prior projects, knowledge production and exchange.

d. Strengthening female participation and voice during Stakeholders Consultations:

- Organize stakeholders' consultations throughout the project cycle. Ensure effective participation of women while conducting consultations to gain knowledge about their specific concerns during project preparation, and to monitor impacts and allow for real-time correction during implementation;
- Use local dialects and involve female facilitators/local women leaders, as well as women representatives from the utility company whenever possible;
Organize women-only discussion groups if appropriate; organize consultations and focus groups at a convenient time for women; and create partnerships with women's groups and nongovernmental organizations (NGOs) to assist in the process and ensure the projects' sustainability.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

Each EE of loan/grant financing under the SRMI Facility will be required to conduct financial management and procurement according to the World Bank policies and procedures.

■ Procurement

Each project supported by the SRMI Facility will follow the World Bank's Procurement Regulations.¹⁵ World Bank procurement staff provide implementation support to the EEs throughout the procurement process. They work with governments to achieve the highest bidding and contract management standards to get the best development results. Each project is developed and supervised by a World Bank task team that includes a procurement specialist. Each project with the exception of the guarantees is also subject to the 'World Bank's Anti-Corruption Guidelines, dated October 15, 2006 and revised in January 2011 and as of July 1, 2016'.¹⁶

■ Financial Management

Each project with the exception of the guarantees supported by the Facility is required to maintain financial management arrangements (including planning and budgeting, accounting, internal control, funds flow, financial reporting, and auditing arrangements of the EE and other entities responsible for project implementation) acceptable to the World Bank. For each project, the task team includes an accredited Financial Management (FM) specialist who provides support during the entire project cycle (preparation, supervision and implementation) in accordance with the Bank's operational policies and procedures (to work with partner countries to design financial management and disbursement arrangements and to supervise and support the performance of Bank-financed operations). The team's FM specialist closely monitors the performance of the EE's FM arrangements and their continued capacity to provide reasonable assurance that financing proceeds are being used for the purposes intended.

Financial reporting will also be required for each project and (with the exception of the guarantees) is specified in each project's Disbursement and Financial Information Letter (DFIL). Unless otherwise agreed by the Bank, the EE must submit annual audited project financial statements six months after the close of the financial year and unaudited interim financial (IFR) reports periodically. Audits are carried out by auditors with independence and capacity acceptable to the Bank, under terms of reference acceptable to the World Bank.

G.4. Disclosure of funding proposal

- ☐ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.
- ☒ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

¹⁵ <https://www.worldbank.org/en/projects-operations/products-and-services/brief/procurement-new-framework#framework-with-the-exception-of-guarantee-projects,-to-which-these-regulations-do-not-apply>.

¹⁶ Guarantees are subject to a similar anti-corruption regime defined in the Anti-Corruption Guidelines for World Bank Guarantee and Carbon Finance Transactions (<https://policies.worldbank.org/sites/ppf3/PPFDocuments/Forms/DispPage.aspx?docid=1971>) and to similar restrictions on participation by debarred entities.

- ☐ full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- ☐ redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

- ☒ Annex 1 NDA no-objection letter(s) [\(template provided\)](#)
- ☒ Annex 2 Feasibility study - and a market study, if applicable
- ☒ Annex 3 Economic and/or financial analyses in spreadsheet format
- ☒ Annex 4 Detailed budget plan [\(template provided\)](#)
- ☒ Annex 5 Implementation timetable including key project/programme milestones [\(template provided\)](#)
- ☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
[\(ESS disclosure form provided\)](#)
 - ☐ Environmental and Social Impact Assessment (ESIA) or
 - ☐ Environmental and Social Management Plan (ESMP) or
 - ☐ Environmental and Social Management System (ESMS)
 - ☒ Environmental and Social Sustainability Framework (ESSF)
- ☒ Annex 7 Summary of consultations and stakeholder engagement plan
- ☒ Annex 8 Gender assessment and project/programme-level action plan [\(template provided\)](#)
- ☒ Annex 9 Legal due diligence (regulation, taxation and insurance)
- ☒ Annex 10 Procurement plan [\(template provided\)](#)
- ☒ Annex 11 Monitoring and evaluation plan [\(template provided\)](#)
- ☒ Annex 12 AE fee request [\(template provided\)](#)
- ☒ Annex 13 Co-financing commitment letter, if applicable [\(template provided\)](#)
- ☒ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

- ☐ Annex 15 Evidence of internal approval [\(template provided\)](#)
- ☐ Annex 16 Map(s) indicating the location of proposed interventions
- ☒ Annex 17 Multi-country project/programme information [\(template provided\)](#)
- ☐ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
- ☐ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☐ Annex 20 First level AML/CFT (KYC) assessment
- ☐ Annex 21 Operations manual (Operations and maintenance)
- ☐ Annex x Other references

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*